

Lecture 05

Application and Impacts of Computing

Chapter lecture — 15 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XII · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — بارہویں جماعت

15 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XII

How to use this BIEK Computer Science lecture

Each PDF is a classroom lecture from the BIEK / Sindh Computer Science textbooks (New Sindh Curriculum, STBB Grade XI–XII). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before looking at the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر بی آئی ای کے / سندھ نصاب کی کتاب کے درس کے مطابق ہے۔ سبق پڑھیں، مثال حل کریں، پھر چیک یورسیلف کے سوالات کے جواب اپنے نوٹ بک میں لکھیں۔ کتاب کے بعد اضافی بورڈ پریکٹس ہے۔ ★ گولڈن ٹاپکس امتحان میں زیادہ پوچھے جاتے ہیں۔

Lectures in this pack

Chapter 5: Application and Impacts of Computing. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 5.1 Machine Learning, Neural Networks & Deep Learning

L02. 5.1.1 Machine Learning (ML)

L03. 5.1.2 Neural Networks (NN)

L04. 5.1.3 Deep Learning (DL)

L05. 5.1.4 Components of Neural Networks

L06. 5.1.5 Components of Deep Learning

L07. 5.2 Applications of Neural Networks and Deep Learning

L08. 5.3 Secure Collaboration (Authentication, Access Control)

L09. 5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

L10. 5.4 Security Threats and Mitigation Techniques

L11. 5.4.2 Identify Security Threats

L12. 5.4.3 Apply Mitigation Techniques

L13. 5.5 Equity and Equal Access in Digital Collaboration

L14. 5.5.1 Functions of Equity and Equal Access

L15. 5.5.2 Collaboration Tools

Application and Impacts of Computing

This chapter shows you how computers learn from data with machine learning, neural networks and deep learning, how to collaborate online without leaking secrets, how to protect data and stop cyber threats, and how to keep digital teamwork fair for every student in Sindh.

اس باب میں آپ سیکھیں گے کہ مشین لرننگ، نیورل نیٹ ورکس اور ڈیپ لرننگ کمپیوٹر کو ڈیٹا سے کیسے سکھاتے ہیں، آن لائن کام کو محفوظ کیسے رکھا جائے، ڈیٹا کی حفاظت اور سائبر خطرات کا مقابلہ کیسے کیا جائے، اور ڈیجیٹل تعاون کو سب کے لیے منصفانہ کیسے بنایا جائے۔

What you will learn

- Define **machine learning (ML)** and describe the three steps: data input, learning, prediction.
- Define **deep learning (DL)** and draw the relationship $AI \supset ML \supset DL$.
- List the six components of deep learning and give applications of NN and DL in daily life and in Pakistan.
- Protect data using encryption (including a Caesar cipher), strong passwords, the 3-2-1 backup rule and a firewall.
- Distinguish **equal access** from **equity** and describe the functions of equity in digital collaboration.
- Explain a **neural network (NN)** as a brain-inspired model with an input layer, hidden layers and an output layer.
- Name the eight components of a neural network and compute a neuron's output from given numbers.
- Explain **secure collaboration**: authentication, multi-factor authentication (MFA) and access control.
- Identify common threats (malware, phishing, DoS) and apply mitigation techniques.
- Name common collaboration tools and use them fairly and safely.

5.1 Machine Learning, Neural Networks & Deep Learning

LEARN IT

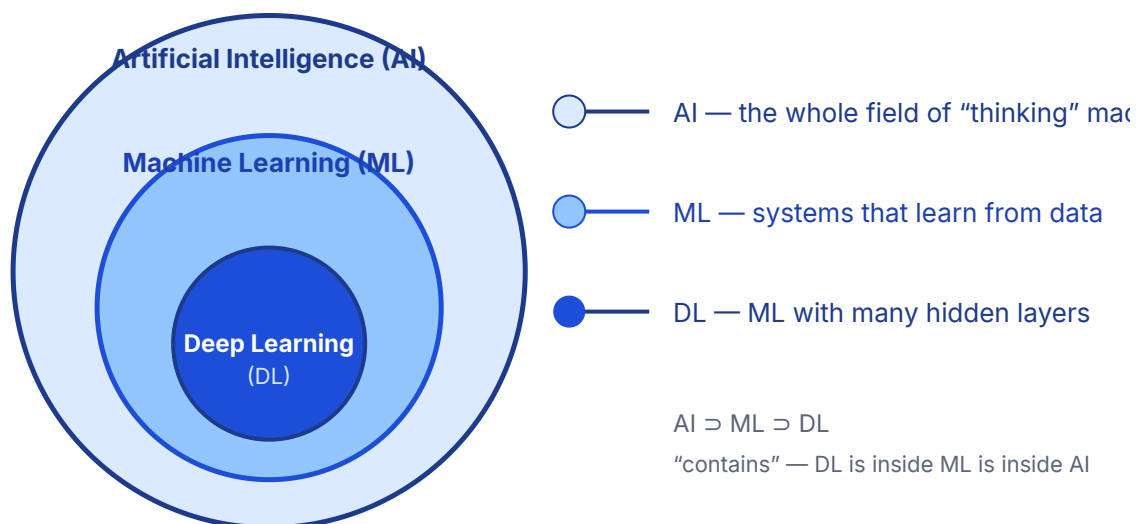
Machine learning (ML), **neural networks (NN)** and **deep learning (DL)** are important areas of **artificial intelligence (AI)**. **AI** is the broad field in which computers perform tasks that normally need human thinking, such as recognising a face, understanding speech or making a decision.

These technologies help computers **learn from data** and make useful decisions **without being explicitly programmed for every single step**. You do not write a separate rule for every possible face or every Urdu accent. You show the computer many examples, and it finds the patterns itself.

They are used in many modern systems:

- **Face unlock** on smartphones.
- **Speech recognition** (for example Siri, Alexa, or Urdu voice typing on a phone).
- **Photo search** categorisation (typing "car" in a gallery and seeing car photos).
- **Medical diagnosis** from X-rays and scans.
- **Language translation** (English to Urdu or Sindhi).

Remember the nesting: **AI is the biggest set**. Inside AI sits **ML**. Inside ML sits **DL**. Every deep-learning system is a kind of machine learning, and every machine-learning system is a kind of AI — but not every AI system is deep learning.



The set relationship $AI \supset ML \supset DL$. Deep learning is a method of machine learning; machine learning is a part of artificial intelligence.

WORKED EXAMPLE

Worked example: place each system in the correct set. Sara in Karachi uses three features on her phone.

1. **A chess app that follows fixed if-then rules** written by a programmer (“if the queen is attacked, move it”). This is **AI** (the machine decides), but it is **not ML**, because it does not learn from data.
2. **A JazzCash fraud filter** that was trained on thousands of past transactions and then flags a new unusual payment. This is **ML** (and therefore also AI). If it uses only a simple model with few layers, it may **not** be DL.
3. **Face unlock** that studied millions of face images through many hidden layers. This is **DL** — and therefore also ML and AI.

Exam sentence you can write: *All DL is ML, all ML is AI, but not all AI is ML, and not all ML is DL.*

REMEMBER IT

Russian-doll memory trick. Picture three boxes. The outer box is **AI**. Inside it is a smaller box **ML**. Inside that is the smallest box **DL**. You cannot take DL out of ML, and you cannot take ML out of AI.

COMMON MISTAKE

Mistake: “AI, ML and DL are three different, separate subjects.”

Fix: They nest. $DL \subset ML \subset AI$. A deep-learning face-unlock system is all three at once.

اردو میں سمجھیں

مصنوعی ذہانت (AI) سب سے بڑا میدان ہے۔ اس کے اندر مشین لرننگ (ML) ہے، اور مشین لرننگ کے اندر ڈیپ لرننگ (DL) ہے۔ یہ ٹیکنالوجیز کمپیوٹر کو ڈیٹا سے سکھاتی ہیں تاکہ ہر قدم الگ الگ پروگرام کیے بغیر فیصلے کیے جا سکیں، جیسے چہرے سے فون کھولنا یا آواز پہچاننا۔

CHECK YOURSELF

1. Which is the largest set: AI, ML or DL?
2. Is every neural-network face-unlock system an example of AI? Why?
3. Give one modern system from this section that uses these technologies.

Answers: 1. AI ($AI \supset ML \supset DL$). 2. Yes — DL is inside ML, and ML is inside AI. 3. Any of: face unlock, speech recognition, photo search, medical diagnosis, language translation.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. ML, NN and DL are parts of:

- a. AI
- b. OSI layer 1 only
- c. Access wizards only
- d. Waterfall only

MCQ 2. They all need:

- a. Data to learn from
- b. Only copper cable
- c. Only primary keys
- d. Only Figma

MCQ 3. The course order is:

- a. ML as the idea, NN as a model, DL as deep NNs
- b. DL then stone tools
- c. SQL then NAND
- d. HCI then K-maps

MCQ 4. Pakistan's AI policy is mentioned as:

- a. A future/local context
- b. A Python library
- c. A chart type
- d. A file mode

MCQ 5. A student mistake is:

- a. Treating the three terms as identical
- b. Giving one example
- c. Mentioning data
- d. Mentioning layers

MCQ 6. This lecture is the map; later ones:

- a. Define each term properly
- b. Delete the chapter
- c. Only careers
- d. Only MVPs

Extra short questions

1. Place ML, NN, DL in one nested sentence.
2. What do they share?
3. What should an XI/XII answer still include?
4. One Pakistani hope from the notes.

Extra long question

1. Draw a nested diagram $AI \supset ML \supset NN \supset DL$ and write one example at each ring.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. DL is a kind of NN which is a way to do ML which is a kind of AI.

2. Learn patterns from data instead of only hand-written rules.
3. A human-responsible, data-quality, bias warning.
4. Agriculture, health, Urdu NLP, public service — any from the book.

Long

1. AI: chess rules engine. ML: spam filter. NN: digit recognition. DL: speech-to-Urdu. One limit at the DL ring (needs lots of data/compute).

5.1.1 Machine Learning (ML)

LEARN IT

A **machine learning model** is the system that **learns from data and gives an output**. Think of it as the **trained brain** of a computer program.

In ordinary programming you write every rule yourself: *if marks ≥ 40 then "Pass"*. In machine learning you **do not list every rule**. You give the model many examples (data), it finds the patterns, and later it decides about new cases it has never seen.

How it works — three steps:

1. Data input → 2. Learning (find patterns) → 3. Prediction on new data

1. **Data input.** The model is given massive amounts of data — for example thousands of past BISE marksheets, or millions of crop photos from Sindh fields.
2. **Learning.** It studies the data and **learns patterns and rules automatically**. It might notice that students who scored low in Chapter 3 also failed the board exam, or that yellow spots on cotton leaves often mean a pest.
3. **Prediction.** After training, it can make **accurate predictions or decisions** when given **new, unseen data** — a new student's record, or a new leaf photograph.

Note from the source: there are many types of machine-learning models. A **neural network** is one important type. **Deep learning** is a highly advanced method of machine learning.

WORKED EXAMPLE

Worked example: a wheat-yield model for a farm near Sukkur.

Step 1 — Data input. For 8 years the farm recorded: rainfall (mm), temperature, fertiliser bags used, and the actual yield (maunds per acre).

Step 2 — Learning. The model studies those rows. It finds, for example, that yield rises with fertiliser up to a point, and falls when temperature is very high in April.

Step 3 — Prediction. This year the farmer types in this season's rainfall and temperature. The model has never seen this exact year before, but it predicts "about 32 maunds per acre." The farmer plans storage and a buyer in advance.

Nobody wrote a giant list of "if rain = 120 and temp = 38 then yield = ...". The model learned the pattern from data.

REMEMBER IT

D-L-P. Machine learning is **Data** in, **Learn** patterns, **Predict**. "Data Learns to Predict."

COMMON MISTAKE

Mistake: "The computer is intelligent on day one."

Fix: An ML model is useful only **after training** on enough good data. Garbage data → garbage predictions. Also, neural networks are *one* type of ML, not the only type; deep learning is a *method* that uses deep neural networks.

اردو میں سمجھیں

مشین لرننگ ماڈل وہ نظام ہے جو ڈیٹا سے سیکھ کر نتیجہ دیتا ہے — کمپیوٹر پروگرام کا تربیت یافتہ دماغ۔ پہلے بہت سا ڈیٹا دیا جاتا ہے، پھر ماڈل خود اصول اور پیٹرن سیکھتا ہے، اور تربیت کے بعد نئے ان دیکھے ڈیٹا پر پیش گوئی کرتا ہے۔ نیورل نیٹ ورک مشین لرننگ کی ایک اہم قسم ہے، اور ڈیپ لرننگ اس کا ایک اعلیٰ طریقہ ہے۔

CHECK YOURSELF

1. What are the three steps of how an ML model works?
2. What is a machine-learning model compared to in the source?
3. After training, what kind of data can the model handle?

Answers: 1. Data input, learning, prediction. 2. The trained brain of a computer program. 3. New, unseen data.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Machine learning lets a model:

- a. Improve from examples, not only hard-coded rules
- b. Run without data
- c. Replace all maths
- d. Ignore errors always

MCQ 2. Training data is:

- a. Examples used to fit the model
- b. The final user password
- c. OSI headers
- d. Figma frames

MCQ 3. A model that memorizes noise is:

- a. Overfitting
- b. Perfect science
- c. A stack
- d. A pie

MCQ 4. Supervised learning uses:

- a. Labels (the right answers)
- b. No data
- c. Only unsupervised always
- d. Only cables

MCQ 5. ML in this book is the “trained brain” metaphor for:

- a. Mapping inputs to outputs after seeing examples
- b. A DBMS trigger
- c. A foreign key
- d. A sprint

MCQ 6. Garbage data yields:

- a. Garbage predictions
- b. Perfect fairness
- c. $O(1)$ time always
- d. Free RAM

Extra short questions

1. Define ML in the book’s words/your own accurate line.
2. Give two applications.
3. What is a label?
4. One ethical issue.

Extra long question

1. Explain ML to a principal who wants to predict drop-outs: data, train, test, a risk, a human check.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A model that learns from data to produce outputs without being explicitly programmed for every case.
2. Spam; recommendations; marks prediction; vision.
3. The known answer attached to a training example.
4. Bias in the training data is copied into decisions.

Long

1. Past attendance+marks → model → test on held-out year. Risk: punishing poverty correlated with absence. Human counsellor decides, model only flags.

5.1.2 Neural Networks (NN) ★ Golden

LEARN IT

A **neural network** is a computer model that **learns patterns from data**. It is **directly inspired by the human brain**, but it works as a **mathematical computer system** — not as living tissue.

Your brain has billions of nerve cells (**neurons**) that pass signals to each other. A neural network copies this idea with small processing units also called **neurons** or **nodes**. These nodes are **connected** and **pass information from one layer to another**, performing calculations at each step.

A typical neural network has three kinds of layer:

- **Input layer** — receives the raw data (pixels of a face, samples of a sound, marks of a student).
- **Hidden layer(s)** — the “thinking” in the middle. Nodes here combine the inputs and learn features (eye distance, nose shape, an edge in a photo).
- **Output layer** — gives the final answer (unlock / keep locked; “this is a cat”; “pass / fail”).

Input layer → Hidden layer(s) → Output layer

WORKED EXAMPLE

Practical example from the source: face unlock.

Step 1 — Input layer. You look at your phone. The camera captures your face as numbers (brightness of each pixel). Those numbers enter the input layer.

Step 2 — Hidden layers. The system studies face features such as **eye distance** and **nose shape**. Early hidden nodes may detect edges and shadows; later hidden nodes combine those into “this looks like Sara’s face.” It compares them with stored data from when you first set up the phone.

Step 3 — Output layer. If the features match, the phone **unlocks**. If they do not (your brother holds the phone), it stays locked.

The same three-layer story works for an NADRA-style biometric gate at a college: camera (input) → feature matching (hidden) → open the turnstile or not (output).

REMEMBER IT

I-H-O. Input receives, Hidden thinks, Output decides. “In Hidden Out.” A neural network is a **brain-inspired maths machine**, not a copy of a living brain.

COMMON MISTAKE

Mistake: "A neural network is an actual human brain inside the computer."

Fix: It is only **inspired by** the brain. It is mathematics: numbers, multiplications and additions running on ordinary hardware.

اردو میں سمجھیں

نیورل نیٹ ورک ایک کمپیوٹر ماڈل ہے جو ڈیٹا سے پیٹرن سیکھتا ہے۔ یہ انسانی دماغ سے متاثر ہے مگر ایک ریاضیاتی نظام کے طور پر کام کرتا ہے۔ چھوٹی اکائیاں نیورون یا نوڈ کہلاتی ہیں، جو تھوں میں جڑی ہوتی ہیں: ان پٹ تھ ڈیٹا لیتی ہے، پوشیدہ تھ سوچتی ہے، اور آؤٹ پٹ تھ فیصلہ دیتی ہے — جیسے چہرے سے فون کھولنا۔

EXAM TIP

Very common questions: "What is a neural network?" (2–3 marks), "Name its three layers and explain each with the face-unlock example" (5 marks), and "How is a neural network related to the human brain?"

Full-mark recipe: (1) definition — computer model that learns patterns from data, inspired by the brain, mathematical; (2) neurons/nodes connected across layers; (3) Input / Hidden / Output, each in one sentence; (4) walk the camera → features (eye distance, nose shape) → unlock path. Draw the three boxes if the paper asks for a diagram.

CHECK YOURSELF

1. What are the small processing units of a neural network called?
2. In face unlock, which layer studies eye distance and nose shape?
3. Is a neural network a living brain? Explain in one sentence.

Answers: 1. Neurons or nodes. 2. Hidden layers. 3. No — it is inspired by the brain but works as a mathematical computer system.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A neural network is inspired by:

- a. The brain's networks of neurons (loosely)
- b. OSI cables
- c. Access forms
- d. Bubble sort

MCQ 2. It learns:

- a. Weights that turn inputs into outputs
- b. SQL indexes
- c. TCP windows
- d. Figma auto-layout

MCQ 3. Simple NNs have:

- a. Input, hidden, output layers
- b. Only a pie
- c. Only one Excel cell
- d. No numbers

MCQ 4. This is ★: define, then:

- a. One diagram + two uses
- b. Only careers
- c. Only MVPs
- d. Only keys

MCQ 5. NN vs linear formula:

- a. NN can stack non-linear layers for complex patterns
- b. NN cannot use data
- c. NN is a queue
- d. NN is HCI only

MCQ 6. Digits on paper are a classic:

- a. NN / vision example
- b. Primary key example
- c. Waterfall example
- d. K-map example

Extra short questions

1. Define neural network.
2. Three layer types.
3. Two applications.
4. Need for data.

Extra long question

1. Draw a 3-layer net for pass/fail from 3 marks. Label input, hidden, output. Explain one training idea in words (weights change). ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A computer model of connected units that learns patterns from data, loosely like neurons.
2. Input, hidden, output.
3. Image recognition; speech; translation.
4. Without enough varied examples the net does not generalise.

Long

1. 3 input nodes, few hidden, 1 output. Training: compare prediction to label, adjust weights. Limit: not a biological brain.

5.1.3 Deep Learning (DL)

LEARN IT

Deep learning is a machine-learning method in which computers use **multi-layer neural networks** to learn **complex patterns** from data.

The word **“deep”** is used because data passes through **several hidden layers** before the system produces the final output. Each layer learns a **different level of detail**:

- **Early layers** learn simple features — edges, shapes, colours.
- **Deeper layers** combine those simple features to understand complex patterns — a whole face, or a specific car model (a Cultus versus a Corolla).

A network with only one or two hidden layers is a **shallow** neural network. Deep learning is always “deep”: it needs **huge amounts of data** and **processing power**.

Neural network vs deep learning

	Neural network (the model)	Deep learning (the method)
What it is	A system/model that has layers, nodes and connections to help it learn.	A specific method that trains neural networks containing many hidden layers .
Depth	Can be simple (shallow) with just one or two hidden layers.	Always “deep,” requiring huge amounts of data and processing power to learn complex patterns.
Role	The architecture (the “machine”).	The way we train a deep version of that machine.

WORKED EXAMPLE

Real-world example from the source: photo gallery search.

A modern gallery app on a smartphone uses deep learning to recognise faces, objects or places in thousands of photos.

Step 1. During training (done by the company, using millions of images) early layers learned edges and colours; deeper layers learned "wheel", "petal", "minaret".

Step 2. The trained network has already scanned your phone's photos and tagged them.

Step 3. You type "**car**" or "**flower**" in the search bar. The app instantly shows related photos because its deep neural network has already learned visual patterns from millions of training images.

A student in Hyderabad can type "Ajrak" or "cricket" and jump to those pictures without scrolling for ten minutes.

REMEMBER IT

Deep = many hidden layers. Shallow NN: 1-2 hidden layers. Deep NN: many hidden layers, huge data, strong computers. Simple features first, complex meaning later.

COMMON MISTAKE

Mistake: "Deep learning and neural networks are two names for the same thing."

Fix: A neural network is the **model** (layers + nodes + connections). Deep learning is the **method** of training networks that have **many** hidden layers. A small NN with one hidden layer is not deep learning.

اردو میں سمجھیں

ڈیپ لرننگ مشین لرننگ کا وہ طریقہ ہے جس میں کئی پوشیدہ تہوں والے نیورل نیٹ ورک استعمال ہوتے ہیں۔ "ڈیپ" اس لیے کہ ڈیٹا کئی تہوں سے گزر کر پیچیدہ پیٹرن سیکھتا ہے: پہلی تہیں کنارے اور رنگ سیکھتی ہیں، گہری تہیں پورا چہرہ یا گاڑی کی قسم پہچانتی ہیں۔ یہ طریقہ بہت سے ڈیٹا اور طاقتور کمپیوٹنگ کا طالب ہے۔

CHECK YOURSELF

1. Why is the method called “deep” learning?
2. What do early layers learn, and what do deeper layers learn?
3. Can a neural network with one hidden layer be called deep learning?

Answers: 1. Because data passes through several hidden layers. 2. Early: simple features (edges, shapes, colours). Deeper: complex patterns (a whole face, a car model). 3. No — that is a shallow network; DL always uses many hidden layers.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Deep learning uses:

- a. Many-layer neural nets
- b. One if-statement
- c. Only Access macros
- d. Only OSI

MCQ 2. "Deep" refers to:

- a. Many hidden layers
- b. Deep sea cables only
- c. Deep stacks in Python only
- d. Deep queues

MCQ 3. DL shines at:

- a. Images, speech, complex patterns
- b. Two-row Excel averages
- c. Primary keys
- d. K-maps of 2 vars

MCQ 4. A cost of DL is:

- a. Lots of data and compute
- b. No data needed
- c. No energy use
- d. Perfect fairness always

MCQ 5. DL is a subset of:

- a. NN / ML / AI
- b. HCI only
- c. SQL only
- d. SDLC only

MCQ 6. Medical imaging in the book is a:

- a. DL application
- b. A foreign key
- c. A pie of NAND
- d. A sprint

Extra short questions

1. Define deep learning.
2. How is it different from a shallow net?
3. Two applications.
4. One Pakistani constraint.

Extra long question

1. Compare ML vs NN vs DL in a 6-row table (definition, depth, data need, example, limit). 10 marks.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. ML using multi-layer neural networks to learn complex patterns.

2. More layers can build higher-level features (edges → shapes → faces).
3. X-ray analysis; voice assistants; translation.
4. GPUs/data/labels are expensive; start with smaller models.

Long

1. Clear nesting. Example row: spam vs digit net vs speech net. Limits: data, bias, black box, energy.

5.1.4 Components of Neural Networks ★ Golden

LEARN IT

The components of a neural network can be understood through its **layered architecture**. Data **enters**, gets **processed**, and **exits**, with mathematical adjustments happening inside.

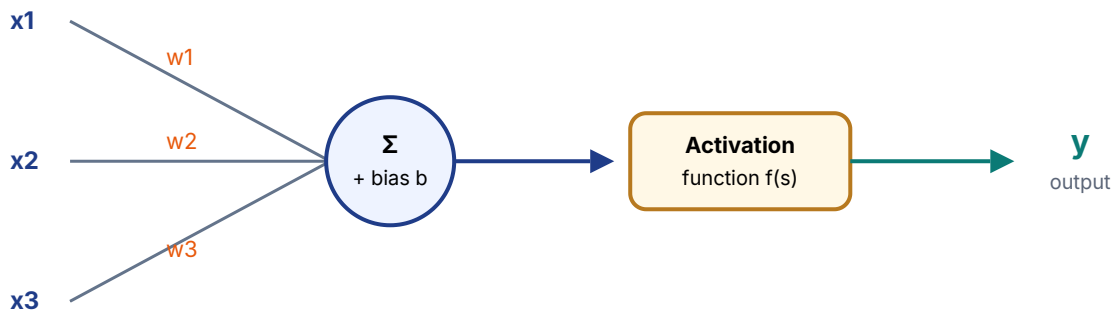
#	Component	Description
1	Input layer	Receives the initial raw data (for example image pixels, sound waves, text).
2	Neurons / nodes	Small processing units that receive, calculate, and pass information forward.
3	Hidden layer(s)	The layers between input and output. They process the data and learn patterns from it.
4	Weights	Numbers that show the importance or strength of connections between neurons.
5	Bias	An extra constant value added to help the model adjust its result and improve accuracy.
6	Activation function	A mathematical gate that decides whether the information is important enough to move to the next layer.
7	Output layer	Gives the final computed result, decision or prediction (for example "This is a cat").
8	Loss function	Checks the error (how much the model's answer is wrong compared with reality) so the network can correct itself.

What happens inside one neuron

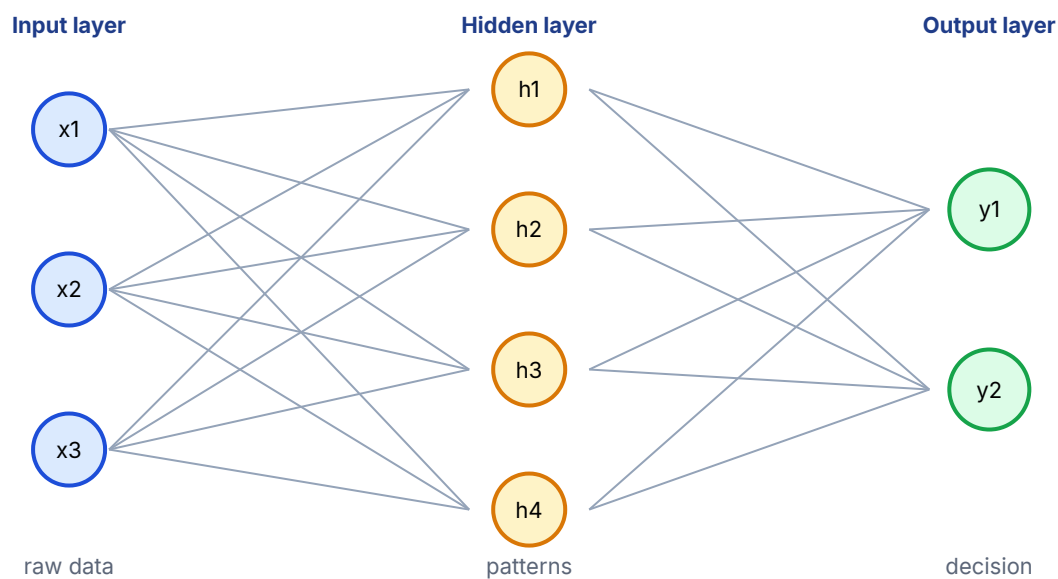
Each neuron does a short calculation:

1. Multiply each input x by its weight w .
2. Add those products together, then add the **bias** b . This total is the **weighted sum** s .
3. Pass s through an **activation function**. A simple exam-friendly gate is a **threshold (step) function**: if s is at least the threshold, the neuron **fires** (output = 1); otherwise output = 0.

In symbols: $s = (x_1 \times w_1) + (x_2 \times w_2) + \dots + b$, then $y = 1$ if $s \geq \text{threshold}$, else $y = 0$.



One neuron: inputs $\times$ weights are summed, bias is added, then an activation function decides the output y .



A small neural network: 3 input nodes, 4 hidden nodes, 2 output nodes. Each line is a weighted connection.

WORKED EXAMPLE

Worked example: neuron calculation with numbers. A college in Larkana uses a tiny network to guess whether a student is ready for a test. One hidden neuron receives two inputs:

Symbol	Meaning	Value
x1	Hours studied yesterday (scaled 0 to 1)	0.8
x2	Attendance this month (scaled 0 to 1)	0.5
w1	Weight of hours studied	0.4
w2	Weight of attendance	0.6
b	Bias	0.1
threshold	Activation gate	0.5

Step 1 — weighted products.

$$x1 \times w1 = 0.8 \times 0.4 = \mathbf{0.32}$$

$$x2 \times w2 = 0.5 \times 0.6 = \mathbf{0.30}$$

Step 2 — add bias to get the weighted sum s.

$$s = 0.32 + 0.30 + 0.1 = \mathbf{0.72}$$

Step 3 — activation (threshold 0.5). Is $0.72 \geq 0.5$? **Yes**, so the neuron fires: **y = 1** (ready for the test).

Step 4 — a second student with $x1 = 0.2$ (almost no study) and $x2 = 0.3$:

$$s = (0.2 \times 0.4) + (0.3 \times 0.6) + 0.1 = 0.08 + 0.18 + 0.1 = \mathbf{0.36}$$

0.36 is **less than** 0.5, so **y = 0** (not ready). The same neuron, different inputs, different decision.

You can check the first student in Python 3:

```
x1, x2 = 0.8, 0.5
w1, w2 = 0.4, 0.6
b = 0.1
s = x1 * w1 + x2 * w2 + b
y = 1 if s >= 0.5 else 0
print(round(s, 2))
print(y)
```

OUTPUT

0.72

1

Loss function, briefly. If the true answer for the first student was actually "not ready" (true $y = 0$) but the model said 1, the loss is high. Training then slightly changes weights and bias so the next guess is closer to reality.

REMEMBER IT

Weights = importance. Bias = extra nudge. Activation = yes/no gate. Loss = how wrong. A large weight means that input matters more. Bias shifts the sum up or down so the gate can open at the right place.

COMMON MISTAKE

Mistake: Forgetting to add the bias, or multiplying instead of adding after the products.

Fix: Always **multiply first** (each x by its w), **then add** the products, **then add b** , **then** apply the activation. Write the four lines in the exam as we did above.

اردو میں سمجھیں

نیورل نیٹ ورک کے اجزاء ہیں: ان پٹ تہ، نیورون، پوشیدہ تہ، وزن (Weights)، بائیس (Bias)، ایکٹیویشن فنکشن، آؤٹ پٹ تہ اور لاس فنکشن۔ ہر نیورون ان پٹ کو وزن سے ضرب دے کر جمع کرتا ہے، بائیس جوڑتا ہے، پھر ایک گیٹ فیصلہ کرتا ہے کہ سگنل آگے جائے یا نہیں۔ وزن تعلق کی طاقت بتاتا ہے؛ لاس فنکشن غلطی ناپتا ہے تاکہ ماڈل خود درست ہو سکے۔

EXAM TIP

Golden questions: "List the components of a neural network" (often 8 names + one line each), "What are weights and bias?", and a **numerical neuron**: given x , w , b and a threshold, find s and y .

Show every arithmetic step. Underline the final y . If asked for a diagram, draw Input → Hidden → Output with circles and connecting lines, and label weights on the lines. Mention the loss function as the component that measures error so the network can correct itself.

CHECK YOURSELF

1. What do weights show?
2. Using $x_1 = 1$, $x_2 = 0$, $w_1 = 0.5$, $w_2 = 0.5$, $b = 0$, threshold = 0.4, what is y ?
3. What does the loss function check?

Answers: 1. The importance or strength of connections between neurons. 2. $s = 1 \times 0.5 + 0 \times 0.5 + 0 = 0.5$; $0.5 \geq 0.4$ so $y = 1$. 3. The error — how much the model's answer is wrong compared with reality.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Input layer:

- a. Receives the features
- b. Always outputs the class
- c. Stores SQL
- d. Draws Figma

MCQ 2. Hidden layers:

- a. Transform features (learned)
- b. Are optional pie charts
- c. Are OSI hubs
- d. Are primary keys

MCQ 3. Output layer:

- a. Produces the prediction
- b. Reads the keyboard
- c. Commits sqlite
- d. Opens VS Code

MCQ 4. Weights are:

- a. Strengths of connections, learned from data
- b. Font weights only
- c. TCP weights
- d. Agile story points

MCQ 5. Activation functions:

- a. Add non-linearity so layers can do more than a straight line
- b. Sort the array
- c. Encrypt the disk
- d. Draw ER

MCQ 6. This is ★: a 5-mark wants:

- a. Named parts + one sentence each + tiny diagram
- b. Only the word "brain"
- c. Only careers
- d. Only MVPs

Extra short questions

1. List components of a NN from the notes.
2. What does a neuron do conceptually?
3. Why more than one hidden layer can help.
4. What is not a component?

Extra long question

1. Draw and label input, 2 hidden, output. Annotate weights on two edges. Explain in 4 sentences how a pass/fail signal moves left to right. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Layers (in/hidden/out), neurons/nodes, weights, activations, (bias).
2. Weighted sum of inputs, then activation.
3. Features of features (deep).
4. A pie chart; a primary key; a sprint.

Long

1. Diagram. Forward pass description. Training mentioned as changing weights, not needed in full backprop maths. One misuse: calling it an exact brain.

5.1.5 Components of Deep Learning

LEARN IT

Deep learning relies on specific advanced components to achieve its high accuracy. These components **differentiate it from basic machine learning**.

#	Component	What it means
1	Large input data	Deep learning is “ data hungry .” It needs massive datasets (thousands or millions of records) for better learning and accuracy.
2	Many hidden layers	Uses dozens or hundreds of hidden layers to learn deeper, more intricate patterns.
3	Automatic feature learning	Unlike older models where humans had to manually highlight important features, deep learning extracts important features by itself .
4	Learning algorithm	Math routines (like backpropagation) that help the system reduce mistakes and improve learning continuously.
5	High computing power	Processing millions of data points across deep layers needs specialised, strong hardware, such as GPUs (Graphics Processing Units).
6	Training process	The model is trained again and again in loops called epochs to gradually improve its final performance.

A **GPU** was originally made for video games’ graphics, but it can do thousands of small multiplications in parallel — exactly what deep networks need. An **epoch** is one full pass through the training data. Ten epochs means the model has seen every training example ten times, adjusting weights a little each time.

Backpropagation means sending the error (from the loss function) **backwards** through the layers so each weight can be nudged in the direction that reduces the next error.

WORKED EXAMPLE

Worked example: cotton-leaf disease detector for a farm in Mirpurkhas. Match each DL component.

1. **Large input data.** Researchers collect 80,000 labelled photos: healthy leaves and leaves with whitefly damage.
2. **Many hidden layers.** The network has 20 hidden layers so it can go from "green edge" to "this exact pest pattern."
3. **Automatic feature learning.** No agronomist types in "look for yellow circles." The network finds the useful spots itself.
4. **Learning algorithm.** After a wrong guess ("healthy" when it was diseased), backpropagation lowers the weights that caused the mistake.
5. **High computing power.** Training runs on a GPU workstation at a university lab in Karachi; a school laptop would take weeks.
6. **Training process.** They train for 30 epochs. After epoch 1 accuracy is 62%; after epoch 30 it is 94% on photos it has not trained on.

REMEMBER IT

D-L-A-L-G-T. Data (lots), Layers (many), Automatic features, Learning algorithm, GPU power, Training epochs. "Deep Learners Always Love Good Training."

COMMON MISTAKE

Mistake: "Deep learning is just 'a neural network'."

Fix: Name the six extras: huge data, many hidden layers, automatic features, a learning algorithm (backpropagation), GPUs, and repeated epochs. A shallow NN with a small spreadsheet is ordinary ML, not DL.

اردو میں سمجھیں

ڈیپ لرننگ کو عام مشین لرننگ سے یہ چیزیں الگ کرتی ہیں: بہت بڑا ڈیٹا، بہت سی پوشیدہ تہیں، خود بخود خصوصیات نکالنا، سیکھنے کا الگورتھم جیسے بیک پروپیگیشن (Backpropagation)، طاقتور ہارڈویئر جیسا کہ جی پی یو (GPU)، اور بار بار تربیت جسے ایپوک (Epoch) کہتے ہیں۔

CHECK YOURSELF

1. Why is deep learning called “data hungry”?
2. What is an epoch?
3. Name the hardware often used for DL, and the learning algorithm named in the source.

Answers: 1. It needs massive datasets (thousands or millions of records) for accuracy. 2. One training loop — one full pass through the training data. 3. GPUs (Graphics Processing Units); backpropagation.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. DL components include:

- a. Deep stacks of layers, lots of data, lots of compute, specialised architectures
- b. A single if
- c. A single Access form
- d. A single NAND

MCQ 2. CNNs are often for:

- a. Images
- b. Undo stacks
- c. Printer queues
- d. Primary keys

MCQ 3. RNNs/transformers often for:

- a. Sequences (text/speech)
- b. Pie charts
- c. ER diamonds
- d. K-map groups

MCQ 4. GPUs help because:

- a. Many multiply-adds in parallel
- b. They store FKs
- c. They draw OSI
- d. They boil water

MCQ 5. Pretrained models let you:

- a. Reuse features (transfer) when you have less data
- b. Skip ethics
- c. Skip labels forever
- d. Skip HCI

MCQ 6. A practical school limit is:

- a. Not enough labelled images/GPU
- b. Too many GPUs always
- c. Too few ethics issues
- d. Too much labelled data always

Extra short questions

1. Name DL-specific components vs a tiny NN.
2. Why not train ImageNet in a college lab from scratch?
3. One architecture name at definition level.
4. Ethics still apply.

Extra long question

1. A team wants DL for Urdu speech marks entry. List DL components they need and three reasons to start smaller (ML or rules).

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Depth, specialised layers (conv), big datasets, accelerators, regularisation.
2. Compute and labelled data; use a pretrained net if anything.
3. CNN for images / transformer for language — no need for full maths.
4. Medical DL that is biased still harms patients.

Long

1. Need: audio data, labels, model, GPU, evaluation. Start smaller: n small, accents, privacy of children's voices, electricity/cost. Human checks the marks.

5.2 Applications of Neural Networks and Deep Learning

LEARN IT

Machine learning, neural networks and deep learning will play a key role in Pakistan's future. The **National AI Policy 2025** highlights how artificial intelligence is now part of national planning, focusing on **innovation, skill-building, ethical use and digital growth**.

Sector	Potential impact in Pakistan
Education	Smart learning systems, online tutoring, automatic assessments, and finding where students need specific help.
Healthcare	Disease detection, medical image analysis, and managing hospital data. Crucial for rural areas lacking specialist doctors.
Agriculture	Monitoring crops, predicting weather, managing irrigation, and estimating yields — highly important for Pakistan's economy.
Business and banking	Fraud detection, customer support (chatbots), market forecasting, and intelligent risk analysis.
Local languages	Speech recognition, translation and text-to-speech tools for Urdu, Sindhi and other regional languages.

To benefit from this future, Pakistan needs skilled individuals. You should focus on **programming (such as Python), mathematics, data analysis, problem-solving** and the **ethical use of technology**.

WORKED EXAMPLE

Worked example: pick a sector and write a three-line “impact paragraph” for 5 marks.

Agriculture: A cotton grower in Sanghar photographs leaves on a cheap phone. A deep-learning model trained on Pakistani crop images flags whitefly early. The farmer sprays only the affected block, saves pesticide cost, and protects yield — which matters because agriculture is a large part of Pakistan's economy.

REMEMBER IT

Five sectors: E-H-A-B-L. Education, Healthcare, Agriculture, Business/banking, Local languages. Policy words: innovation, skills, ethics, digital growth.

COMMON MISTAKE

Mistake: Writing only “AI will make Pakistan modern” with no sector or example.

Fix: Name the National AI Policy 2025, pick two sectors from the table, and give a local example (rural hospital without a specialist; Urdu/Sindhi speech tools; bank fraud).

اردو میں سمجھیں

قومی اے آئی پالیسی 2025 کے مطابق مصنوعی ذہانت پاکستان کی منصوبہ بندی کا حصہ ہے: جدت، مہارت، اخلاقی استعمال اور ڈیجیٹل ترقی۔ تعلیم، صحت، زراعت، بینکنگ اور اردو و سندھی جیسی مقامی زبانوں میں اس کے اثرات ہوں گے۔ طلبہ کو پائتھن، ریاضی، ڈیٹا کا تجزیہ، مسئلہ حل کرنا اور ٹیکنالوجی کا اخلاقی استعمال سیکھنا چاہیے۔

CHECK YOURSELF

1. Name the national policy mentioned for AI in Pakistan.
2. Why is AI in healthcare especially important for rural areas?
3. List three skills students should learn to prepare for this future.

Answers: 1. National AI Policy 2025. 2. Those areas often lack specialist doctors; AI can help with disease detection and medical image analysis. 3. Any three of: programming (Python), mathematics, data analysis, problem-solving, ethical use of technology.

LEARN IT

Neural networks and deep learning are widely used in modern life. Many intelligent systems use them in the background to **recognise information, make predictions and support users**.

#	Application	What it does	Example
1	Image recognition	Identifies objects, faces or patterns in images. Used in security cameras, photo search and biometric locks.	Unlocking a smartphone using face recognition; a camera at a Karachi plaza matching a face with a staff list.

2	Speech recognition	Converts spoken words into digital text or commands. Deep learning helps with accents, pronunciation and voice patterns.	A student speaks to Google Assistant or Alexa to set a morning alarm; Urdu voice search on a phone.
3	Natural Language Processing (NLP)	Helps computers understand, process and respond to human language grammatically and in context. Core of chatbots, translators and AI writers.	Using Google Translate to turn a difficult English paragraph into Urdu; a bank chatbot answering in Urdu.
4	Healthcare systems	Analyse complex medical images (X-rays, CT scans, MRIs) to detect diseases early and accurately.	A hospital system highlighting abnormal patterns in a lung X-ray that might indicate pneumonia or tumours — useful at a DHQ hospital that has few radiologists.
5	Recommendation systems	Analyse past behaviour and preferences to suggest content, products or services.	A student watches computer-science videos on YouTube; the platform recommends more programming tutorials.
6	Autonomous vehicles	Self-driving cars use neural networks to process real-time camera and radar data: roads, signals, pedestrians, obstacles.	A self-driving car detects a pedestrian crossing and automatically applies the brakes.
7	Fraud detection	Banks compare current activity with previous spending patterns to catch unusual transactions.	An HBL or Meezan app flags a sudden Rs 80,000 payment from a new city at 3 a.m.

8	Assistive technologies	AI supports people with disabilities.	Speech-to-text tools for users who have difficulty typing; screen readers that speak Urdu text aloud.
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WORKED EXAMPLE

Worked example: match the app to the application name — a day in Ahmed's life in Hyderabad.

1. 7:00 am — phone unlocks by face. → **Image recognition.**
2. 7:05 am — "Ok Google, set alarm for Maghrib." → **Speech recognition.**
3. 8:00 am — Google Translate helps with an English article. → **NLP.**
4. 4:00 pm — YouTube suggests a Python tutorial after he watched Chapter 5 notes. → **Recommendation system.**
5. 9:00 pm — JazzCash texts "We blocked a payment that does not match your pattern." → **Fraud detection.**

REMEMBER IT

Eight applications: I-S-N-H-R-A-F-A. Image, Speech, NLP, Healthcare, Recommendations, Autonomous vehicles, Fraud, Assistive tech. "I See New Hospitals Recommend Autonomous Fraud-Assist."

COMMON MISTAKE

Mistake: Mixing up speech recognition and NLP.

Fix: Speech recognition turns **sound into text** (or a command). NLP understands the **meaning of language** (grammar, context, translation, chat replies). You can have NLP on typed text with no microphone at all.

اردو میں سمجھیں

نیورل نیٹ ورکس اور ڈیپ لرننگ کی مشہور ایپلی کیشنز یہ ہیں: تصویر کی پہچان، آواز کی پہچان، قدرتی زبان کی پروسیسنگ (NLP)، صحت کے نظام، تجویز کرنے والے نظام، خود چلنے والی گاڑیاں، فراڈ کی نشاندہی اور معاون ٹیکنالوجیز۔ روزمرہ میں چہرے سے فون کھولنا، یوٹیوب کی تجاویز اور بینک کا مشکوک لین دین پکڑنا سب اسی خاندان سے ہیں۔

CHECK YOURSELF

1. Which application converts spoken words into text or commands?
2. Give the source example of a recommendation system.
3. How do self-driving cars use neural networks?

Answers: 1. Speech recognition. 2. YouTube recommending more programming tutorials after a student watches computer-science videos. 3. They process real-time camera and radar data to understand roads, traffic signals, pedestrians and obstacles (for example braking for a pedestrian).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Vision applications include:

- a. Face unlock, medical images, number-plate reads
- b. Primary keys
- c. Bubble sort
- d. OSI ping

MCQ 2. Language applications include:

- a. Translation, speech-to-text, chatbots
- b. NAND grouping
- c. Foreign keys
- d. Pie charts of cables

MCQ 3. Healthcare DL must still:

- a. Be checked by clinicians
- b. Replace doctors unsupervised always
- c. Ignore bias
- d. Train on one patient

MCQ 4. A local example could be:

- a. Urdu voice assistants / crop images
- b. K-maps of farms
- c. Access reports of OSI
- d. Figma of NAND

MCQ 5. Recommendation systems are:

- a. ML/NN apps (what to watch/buy)
- b. Queues in printers only
- c. Stacks of plates only
- d. ER only

MCQ 6. The exam wants:

- a. Named field + how NN/DL helps + one risk
- b. Only the word Pakistan
- c. Only careers list
- d. Only math of backprop

Extra short questions

1. Four application areas from the book.
2. One risk in health.
3. One risk in security/surveillance.
4. Why "application" questions still need a definition line.

Extra long question

1. Pick vision, health, language. For each: system, data it needs, a benefit, a harm. 10-mark table.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Vision, speech, health, transport, security, education — any four with a concrete system.
2. False negative on an X-ray; bias against a skin tone.
3. Misidentification; privacy.
4. 1 mark for the term, rest for use+limit.

Long

1. Face gate / images / fast entry / racism in scores. X-ray / labelled scans / earlier catch / missed cases. Urdu STT / audio / inclusion / leaking class recordings.

5.3 Secure Collaboration (Authentication, Access Control)

★ Golden

LEARN IT

Digital tools allow people to **share files, attend online classes, edit documents and work together from different locations**. Collaboration is useful only when **data is protected** and **communication remains safe**.

Secure collaboration means using digital tools to work together in a **safe, controlled and responsible** way. It depends on several key elements.

i) Authentication / identity verification

Authentication is **confirming the identity of a user before allowing access**. Common methods:

- **Username and password**
- **PIN** (a short number code)
- **OTP — One-Time Password**, a code that works only once, usually sent by SMS or an authenticator app
- **Biometrics** — fingerprint or face

Source example: a **banking app sending an OTP to your phone before login**.

Multi-factor authentication (MFA) and two-factor authentication (2FA)

A **factor** is a category of proof:

- **Something you know** — password or PIN.
- **Something you have** — your phone, an OTP, a bank SMS.
- **Something you are** — fingerprint, face (biometrics).

Multi-factor authentication (MFA) means the system asks for **two or more different factors** before it trusts you. **Two-factor authentication (2FA)** is the common case of MFA that uses **exactly two** factors — usually password + OTP.

If a thief in another city steals only your password, MFA still blocks them because they do not have your phone (or your fingerprint).

ii) Access control

Access control decides **what an authenticated user is allowed to do**: View, Edit, Comment, Download, Share.

Authentication answers **"Who are you?"** Access control answers **"What may you do?"**

Source example: a teacher shares a Google Doc with students as **"view only"**, but gives group leaders **"edit"** permissions.

iii) Accountability and tracking

User actions are linked to **verified accounts**. You can see who opened, edited, deleted or shared a file. Example: a school **LMS** (Learning Management System) tracks which teacher uploaded an assignment and which student submitted it.

iv) Data privacy

Data privacy means protecting personal and sensitive information and using it **only for the purpose it was collected**. It involves:

- **Protection** — keeping names, numbers, marks and passwords secure.
- **Controlled sharing** — sharing only with the right people.
- **Confidentiality** — data stays private during storage and transmission.
- **Compliance** — following legal and ethical rules for data handling.

v) Safe communication / transmission

Exchanging information through digital platforms **securely**, protecting messages from unauthorised interception or wiretapping. Example: a teacher shares class instructions through an **official, encrypted LMS** rather than an open public forum or an unknown WhatsApp broadcast.

WORKED EXAMPLE

Worked example: a Class XII Google Classroom in Sukkur.

Step 1 — Authentication. Each student signs in with a school email and password. The college turns on MFA: after the password, Google sends an OTP to the student's registered Jazz/Zong number.

Step 2 — Access control. All 40 students can **view** the assignment sheet. Only the four group leaders can **edit** the shared report. Nobody except the teacher can **delete** the class materials. A student who left the college is removed from the class (access revoked).

Step 3 — Accountability. Version history shows that Ali edited paragraph 2 at 8:14 pm and that Hina submitted the PDF at 9:02 pm.

Step 4 — Privacy and safe transmission. Marks stay inside Classroom; the teacher does not post CNIC numbers or phone numbers in a public Facebook group. Traffic to Google is encrypted, unlike an open public forum.

If Ali shares his password with a cousin, authentication is broken. If the teacher gives every student "edit" on the marksheet, access control is broken even though login still works.

REMEMBER IT

A-A-A-P-S. Authentication, Access control, Accountability, Privacy, Safe transmission.
MFA = at least two different factor types (know / have / are). 2FA is MFA with two factors.

COMMON MISTAKE

Mistake: Treating authentication and access control as the same thing.

Fix: Login only proves identity. After login, access control still limits View / Edit / Comment / Download / Share. A student who is authenticated must not automatically get teacher rights.

اردو میں سمجھیں

محفوظ تعاون کا مطلب ہے ڈیجیٹل ٹولز سے محفوظ، کنٹرول شدہ اور ذمہ دارانہ انداز میں ساتھ کام کرنا۔ تصدیق (Authentication) یہ جانچتی ہے کہ صارف کون ہے؛ رسائی کنٹرول (Access Control) یہ طے کرتا ہے کہ وہ کیا کر سکتا ہے (صرف دیکھنا، ترمیم، تبصرہ)۔ کثیر عنصری تصدیق (MFA) دو یا زیادہ مختلف ثبوت مانگتی ہے، جیسے پاس ورڈ اور او ٹی پی (OTP)۔ حساب دہی، ڈیٹا پرائیویسی اور محفوظ ترسیل بھی ضروری ہیں۔

EXAM TIP

Typical items: "Define secure collaboration", "Differentiate authentication and access control", "What is MFA/2FA?", "Explain five elements of secure collaboration with examples."

Write the five headings (authentication, access control, accountability, privacy, safe transmission). For MFA, name the three factor types and say that 2FA is password + OTP in a banking app. Use the Google Doc "view only vs edit" example — examiners look for it.

CHECK YOURSELF

1. What question does authentication answer, and what question does access control answer?
2. Password + OTP is an example of which idea?
3. Give the source example of access control in a Google Doc.

Answers: 1. Authentication: who are you? Access control: what may you do? 2. Two-factor / multi-factor authentication (2FA/MFA). 3. Teacher shares the document as view-only with students but gives group leaders edit permission.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Authentication is:

- a. Proving who you are
- b. What you are allowed to do
- c. Encrypting a disk
- d. A pie chart

MCQ 2. MFA means:

- a. More than one factor (password + OTP/phone)
- b. Many files always
- c. Many Figma frames
- d. Many OSI layers

MCQ 3. Access control is:

- a. What an authenticated user may do
- b. The login spelling
- c. A histogram
- d. A stack

MCQ 4. Accountability needs:

- a. Actions tied to accounts (logs)
- b. Shared "class" password
- c. No names
- d. Disabled logs

MCQ 5. A shared Google-doc with "anyone can edit" fails:

- a. Access control / accountability
- b. Big O
- c. NAND
- d. Median

MCQ 6. This is ★: define collaboration risk, then:

- a. Auth, MFA, access control, tracking
- b. Only careers
- c. Only MVPs
- d. Only charts

Extra short questions

1. Define authentication vs access control.
2. Two MFA factors.
3. Why logs?
4. A class example of bad practice.

Extra long question

1. Design login for a college LMS: password rules, MFA, roles (student/teacher/admin), and what an audit log stores. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Auth: who. Access: what they can do.
2. Something you know (password) + have (phone OTP) / are (fingerprint).
3. After a leak you can see who opened the file.
4. One Facebook password on the projector.

Long

1. Roles table. Students cannot edit marks. Teachers cannot drop the DB. MFA for admin.
Logs: user, time, action, IP. No shared accounts.

5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

LEARN IT

Data protection means keeping digital information safe from **unauthorised access, misuse, loss, damage or accidental deletion**.

1. Data encryption

Encryption changes readable data (**plaintext**) into a coded form called **ciphertext**.

Unauthorised users cannot understand ciphertext without the key.

- **Symmetric encryption** — the **same key** is used to encrypt and to decrypt.
- **Asymmetric encryption** — **two keys**: a **public key** to encrypt, a **private key** to decrypt.

A simple exam cipher is the **Caesar cipher** (a symmetric substitution cipher). You shift every letter a fixed number of places along the alphabet. Shift 3 is the classic key: A → D, B → E, ..., Z → C (the alphabet wraps around).

2. Strong passwords

A strong password includes **uppercase, lowercase, numbers and symbols**. It resists **brute-force** guessing (trying every possible combination). Source example: `Zk@2026#Pass`.

Weak: `ahmed123`, `pakistan`, `sindh`, your date of birth. Never reuse the same password on the college LMS and on JazzCash.

3. Backup and recovery — the 3-2-1 rule

A **backup** is a copy of data in a safe location (cloud or external drive) so you can **restore** it after hardware damage, ransomware or accidental deletion.

The **3-2-1 backup rule** is a practical standard:

- **3** copies of the data (the original plus two backups).
- **2** different types of storage media (for example laptop disk + USB, or disk + cloud).
- **1** copy **off-site** (a different building or the cloud), so a fire or theft at home does not destroy every copy.

4. Firewall and antivirus software

A **firewall** is a security **barrier** that **monitors network traffic** and **blocks suspicious internet access**. Think of it as a checkpoint on the road between your device (or college lab) and the internet.

Antivirus software detects and removes harmful software (viruses, spyware and other malware) from the **local machine**.

Firewall ≈ guards the **door to the network**. Antivirus ≈ searches **inside the house** for intruders already present.

WORKED EXAMPLE

Worked example A — Caesar cipher, shift = 3. Encrypt the plaintext **SINDH**, then decrypt the result.

Write the alphabet and count three letters forward. Wrap after Z back to A.

Plain	S	I	N	D	H
+3	V	L	Q	G	K

Ciphertext = **VLQ GK**.

Decrypt: shift each letter **back** by 3: V → S, L → I, Q → N, G → D, K → H. You recover **SINDH**.

Spaces and punctuation are often left unchanged. The same key (3) is used both ways, so Caesar is **symmetric**.

Second check: **CAT** + 3 → C→F, A→D, T→W → **FDW**.

```
def caesar(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord("A") if ch.isupper() else ord("a")
            result += chr((ord(ch) - base + shift) % 26 + base)
        else:
            result += ch
    return result

print(caesar("SINDH", 3))
print(caesar("VLQ GK", -3))
print(caesar("CAT", 3))
```

OUTPUT

```
VLQ GK
SINDH
FDW
```

Worked example B — 3-2-1 for Fatima's Class XII project in Hyderabad.

1. **Copy 1 (original):** project folder on her laptop hard disk.
2. **Copy 2 (different media):** the same folder on a USB drive she keeps in her bag.
3. **Copy 3 (off-site):** uploaded to Google Drive (cloud servers are not in her house).

That is 3 copies, 2 media types (internal disk + USB, with cloud as the off-site copy), 1 off-site. When her brother deletes the laptop folder, she restores from USB or Drive.

Password check: Zk@2026#Pass has uppercase Z, lowercase letters, digits 2026, and symbols @ and #. fatima has none of those extras — reject it.

3 copies



2 media types



1 copy off-site

REMEMBER IT

Cipher = lock the message. Password = lock the door. 3-2-1 = spare keys in three places. Firewall = gate. Antivirus = indoor search. Caesar: shift letters; same key both ways (symmetric).

COMMON MISTAKE

Mistake 1: Shifting the wrong way, or forgetting wrap-around (X, Y, Z + 3 → A, B, C).

Mistake 2: Two copies on the same laptop (Documents and Desktop) and calling it 3-2-1. That is still one disk in one house.

Fix: Count letters on a written alphabet. For backups, different media and one copy that would survive a house fire.

اردو میں سمجھیں

ڈیٹا پروٹیکشن کا مطلب ہے معلومات کو غیر مجاز رسائی، غلط استعمال، نقصان یا حادثاتی حذف سے بچانا۔ انکرپشن پڑھے جانے والے متن کو سائفر ٹیکسٹ بنا دیتی ہے؛ سیزر سائفر میں ہر حرف کو مقررہ جگہوں سے آگے سرکایا جاتا ہے۔ مضبوط پاس ورڈ میں بڑے، چھوٹے حروف، عدد اور علامات ہوں۔ 1-2-3 بیک اپ: تین کاپیاں، دو قسم کے میڈیا، ایک کاپی باہر (آف سائٹ)۔ فائر وال نیٹ ورک ٹریفک روکتی ہے؛ اینٹی وائرس مقامی نقصان دہ سافٹ ویئر ہٹاتا ہے۔

CHECK YOURSELF

1. Encrypt PAK with a Caesar shift of 3.
2. What do the 3, the 2 and the 1 stand for in 3-2-1 backup?
3. In asymmetric encryption, which key encrypts and which key decrypts?

Answers: 1. SDN (P→S, A→D, K→N). 2. 3 copies, 2 different media, 1 off-site copy. 3. Public key encrypts; private key decrypts.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Encryption makes data:

- a. Unreadable without a key
- b. Faster to sort
- c. A primary key
- d. A pie

MCQ 2. A strong password is:

- a. Long, unique, not a word+123
- b. Ali123 for every site
- c. Written on the monitor
- d. OTP posted in the group

MCQ 3. Backup is:

- a. A second copy you can restore
- b. A stronger GPU
- c. A prettier UI
- d. A faster sort

MCQ 4. A firewall:

- a. Filters traffic in/out
- b. Encrypts Excel cells one-by-one always
- c. Sorts arrays
- d. Draws ER

MCQ 5. Ransomware defence includes:

- a. Offline/offline-ish backups
- b. Paying and hoping always
- c. No updates
- d. Shared admin

MCQ 6. Data protection is about:

- a. Confidentiality, integrity, availability (CIA idea)
- b. Only colours
- c. Only NAND
- d. Only sprints

Extra short questions

1. Four protection techniques from the notes.
2. Why unique passwords?
3. 3-2-1 backup idea (simple).
4. Firewall vs antivirus in one line.

Extra long question

1. A college lab is hit by ransomware. Using encryption, passwords, backup, firewall: what should already have been in place, and what do you do now (without paying if possible)?

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Encryption, strong passwords, backup, firewall (and updates, least privilege).
2. One leaked site should not open email+JazzCash.
3. More than one copy, not only on the same laptop.
4. Firewall: traffic gate. AV: looks for malware on the host (both useful).

Long

1. Before: backups offline, least admin, updates, firewall. Now: isolate PCs, restore from backup, change passwords, report. Paying is not a syllabus requirement to endorse.

5.4 Security Threats and Mitigation Techniques ★ Golden

LEARN IT

A **security threat** is any action, event or weakness that can **harm a computer system or its data**. **Mitigation techniques** are protective measures used to prevent threats or reduce their impact. This section names the common threats; later sections show how to spot them and how to reduce the damage.

#	Threat	What it is	Example
1	Malware	Harmful software designed to damage a computer, steal data, or gain unauthorised access. Includes viruses, worms, spyware, ransomware and Trojans .	Installing cracked software that secretly corrupts files, or a "free game" USB that encrypts a college lab's coursework (ransomware).
2	Phishing	A social-engineering trick that deceives users into sharing personal information (passwords, bank details, OTPs) using fake emails or websites that look genuine.	An SMS that looks like JazzCash or HBL asking you to "verify" by typing your OTP on a fake page.
3	Unauthorised access	Entering a system, account or file without permission. Often caused by weak passwords, shared passwords, or leaving devices unlocked.	A classmate using an unlocked lab PC to open someone else's Google Classroom.

4	Data theft / identity theft	Data theft is stealing or copying data without permission. Identity theft is using another person's stolen details (name, CNIC, bank details) to pretend to be them online and commit fraud.	A leaked list of CNICs used to open a SIM or a wallet in someone else's name.
5	Denial of Service (DoS)	A website or system is flooded with thousands of fake requests, causing it to slow down or crash, so real users cannot access the service.	Hackers flooding an online exam portal so students cannot log in on BISE result day or during an e-test.
6	Human error	Mistakes by legitimate users: accidentally deleting files, sending confidential data to the wrong email, or falling for a simple scam.	A teacher emailing the whole marksheet to the wrong parent; a student forwarding an OTP "to help the bank."

Malware types in one line each: a **virus** attaches to files and spreads when you share them; a **worm** spreads by itself across a network; **spyware** secretly watches you; **ransomware** locks your files and demands money; a **Trojan** pretends to be useful software but hides harm.

Social engineering means tricking a **person**, not breaking a lock with maths. Phishing is the most common exam example.

Phishing
Fake bank email / SMS
steals password or OTP

DoS attack
Flood of fake requests
server overload / crash

Identity theft
Stolen CNIC / details
fraud in your name

Three high-yield threats: phishing (fake messages), DoS (server overload), identity theft (stolen CNIC or bank details).

WORKED EXAMPLE

Worked example: classify the incident.

1. Bilal downloads a "cracked" Photoshop. Next day his project files are unreadable and a note asks for Bitcoin. → **Malware** (ransomware), started by cracked software.
2. Ayesha gets an email "Your Meezan account will close in 2 hours. Click here and send OTP." The link is meezn-pk.tk, not the real bank. → **Phishing**.
3. On e-test morning the college portal spins forever; the admin graph shows 200,000 requests from one pattern of fake traffic. → **DoS**.
4. Someone opens a JazzCash wallet using Ahmed's CNIC copy from a leaked form. → **Identity theft** (after data theft).
5. The lab assistant emails the password list to the whole staff by choosing the wrong group. → **Human error**.

REMEMBER IT

Malware = bad program. Phishing = fake message. DoS = flood. Identity theft = stolen 'you'. Human error = honest mistake. Cracked software and unknown USBs are the usual door for malware in Pakistani colleges.

COMMON MISTAKE

Mistake: Calling every cyber problem a "virus", or mixing phishing with malware.

Fix: A virus is only one kind of malware. Phishing is a **trick against the user**; it may not install any file at all. DoS does not steal your password — it makes the service unusable for everyone.

اردو میں سمجھیں

سیکیورٹی خطرہ وہ عمل یا کمزوری ہے جو نظام یا ڈیٹا کو نقصان پہنچاے۔ میلویئر نقصان دہ سافٹ ویئر ہے (وائرس، ورم، سپائی ویئر، رینسم ویئر، ٹروجن)۔ فٹنگ جعلی ای میل یا ویب سائٹ سے پاس ورڈ اور او ٹی پی چھینتی ہے۔ ڈو ایس (DoS) جعلی درخواستوں کی بارش سے سائٹ کریش کر دیتا ہے۔ دیگر خطرات: غیر مجاز رسائی، ڈیٹا/شناخت کی چوری، اور انسانی غلطی۔

EXAM TIP

Expect "Define a security threat", "Explain malware / phishing / DoS with examples", and "Differentiate data theft and identity theft." List all six threats if the question says "common security threats." For DoS, write: flood of fake requests → slowdown or crash → real users blocked, and quote the online exam portal. For phishing, write social engineering + fake genuine-looking email/site + passwords/OTPs.

CHECK YOURSELF

1. Name five types of malware listed in the source.
2. What does a DoS attack do to a website?
3. How is identity theft different from data theft?

Answers: 1. Viruses, worms, spyware, ransomware, Trojans. 2. Floods it with thousands of fake requests so it slows down or crashes and real users cannot access it. 3. Data theft is stealing/copying data; identity theft is using stolen personal details (name, CNIC, bank) to pretend to be that person and commit fraud.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Malware is:

- a. Malicious software (virus, worm, trojan, ransomware...)
- b. A friendly backup
- c. A primary key
- d. A caption

MCQ 2. Phishing is:

- a. Tricking you into giving secrets (fake mail/site)
- b. A fishing sport in the syllabus
- c. A sort
- d. A chart

MCQ 3. DoS aims to:

- a. Make a service unavailable by flooding it
- b. Encrypt with your key
- c. Fix UX
- d. Normalise tables

MCQ 4. A threat is:

- a. Anything that can harm system or data
- b. Only earthquakes
- c. Only NAND delay
- d. Only slow Wi-Fi in exams

MCQ 5. This is ★: name, one example, one mitigation.

- a. The 3-mark recipe
- b. Only the name
- c. Only Python
- d. Only Figma

MCQ 6. A fake BIEK result link is likely:

- a. Phishing
- b. A K-map
- c. A stack
- d. A median

Extra short questions

1. Define malware, phishing, DoS.
2. One extra threat from the notes (theft/identity).
3. A student-level mitigation for phishing.
4. Why USB etiquette?

Extra long question

1. Table of four threats: what it is, a Sindh-college example, a sign, a mitigation. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Malware: bad software. Phishing: social trick. DoS: flood to outage.
2. Data/identity theft — using someone's CNIC/account.
3. Check URL, don't tap unknown SMS, official portal only.
4. Worms travel on sticks.

Long

1. Malware, phishing, DoS, theft. Signs: slowness, fake from-addresses, site down, unknown logins. Mitigations: updates/AV, URL check, backups+capacity, MFA.

5.4.2 Identify Security Threats

LEARN IT

Identifying security threats means **noticing signs** that data or systems may be at risk. Early identification helps you act before serious damage occurs.

Sign / indicator	What it likely means
Slow performance, missing files, or random pop-ups.	Malware or background unauthorised activity.
Failed login attempts or strange login locations.	Attempted unauthorised access .
Emails urging quick action, asking for an OTP, or containing bad grammar / strange links.	Classic phishing .
Unexpected deletion, copying or movement of records.	Data theft or insider misuse.
Sudden massive spike in server requests or a very slow website.	An ongoing DoS attack.

The first line of defence is also simple behaviour: avoid **unsafe downloads, cracked software, unverified websites and infected USB drives**.

WORKED EXAMPLE

Worked example: read the signs in a college computer lab in Khairpur.

1. Three PCs show pop-ups and Word files have vanished after students used a "game" USB. → Treat as **malware**. Disconnect from the network; run antivirus; stop USB use.
2. Google Classroom emails: "Your account will be deleted in 30 minutes. Confirm password here." Link spelling is go0gle. → **Phishing**. Do not click; report to the teacher.
3. The e-portal that was fine at 8:00 am is unreachable at 8:05; the technician sees a sudden spike in requests. → Suspect **DoS**.
4. Nadia's Gmail security page shows a login from another country at 4 a.m. → **Unauthorised access** attempt. Change password; turn on MFA.

REMEMBER IT

Slow/pop-ups → malware. Odd logins → unauthorised access. Urgent OTP email → phishing. Files moving → theft. Traffic spike → DoS.

COMMON MISTAKE

Mistake: Ignoring early signs ("the PC is just old") until the ransomware note appears.

Fix: One unusual login or one OTP email is enough to investigate. Early action is cheaper than recovery.

اردو میں سمجھیں

خطرے کی جلد پہچان نقصان سے پہلے اقدام ممکن بناتی ہے۔ سست روی، غائب فائلیں یا پاپ اپ میلویئر کی علامت ہیں؛ عجیب لاگ ان غیر مجاز رسائی کی؛ فوری او ٹی پی والی ای میل فہشنگ کی؛ ریکارڈ کا اچانک منتقل ہونا ڈیٹا چوری کی؛ سرور پر درخواستوں کا دھماکہ ڈو ایس کی۔ کریک سافٹ ویئر، نامعلوم یو ایس بی اور غیر تصدیق شدہ ویب سائٹوں سے بچیں۔

CHECK YOURSELF

1. Random pop-ups and missing files most likely indicate what?
2. An email with bad grammar that demands your OTP is a sign of what?
3. Name four unsafe habits that invite these threats.

Answers: 1. Malware (or background unauthorised activity). 2. Phishing. 3. Unsafe downloads, cracked software, unverified websites, infected USB drives.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

MCQ 2. VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

MCQ 3. The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

MCQ 4. Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

MCQ 5. A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

MCQ 6. Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Integrated Development Environment: editor + run + debug + files in one place.

2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

Long

1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.

5.4.3 Apply Mitigation Techniques

LEARN IT

Mitigation techniques are actions taken to **prevent** security threats or **reduce their harmful effects** once identified. Technology helps, but it cannot stop all threats if the user is careless.

#	Technique	How it helps
1	Antivirus and firewalls	Antivirus scans and removes harmful software (for example from USB drives). A firewall monitors network traffic and blocks suspicious access from the internet.
2	Strong passwords and 2FA	Complex passwords slow brute-force guessing. 2FA adds an extra layer: an OTP on your phone as well as the password.
3	Data encryption	Converts data into unreadable code during transmission. Protects passwords and sensitive data if they are intercepted.
4	Regular backups	Copies on cloud or external drives allow quick recovery after a crash, human error or ransomware.
5	User awareness (the human factor)	Training users to avoid unknown links, suspicious attachments and password sharing. The human is often the weakest link — and the strongest defence when trained.

WORKED EXAMPLE

Worked example: match each threat to a mitigation — BISE e-test week at a college in Larkana.

Threat seen	Mitigation to apply
USB games in the lab, pop-ups starting.	Scan with antivirus ; block unknown USBs; enable the lab firewall .
Phishing SMS about "update your portal password."	User awareness : do not click; verify on the official site typed by hand. Turn on 2FA .
Fear that intercepted Wi-Fi could show passwords.	Use sites with encryption (HTTPS / encrypted LMS); never send passwords in plain WhatsApp.
Ransomware risk to project files.	Regular 3-2-1 backups so files can be restored without paying.
Shared passwords on a sticky note on the monitor.	Strong unique passwords + 2FA; remove the note; awareness session.

REMEMBER IT

Five shields: A-F, P-2, E, B, U. Antivirus+Firewall, Passwords+2FA, Encryption, Backups, User awareness. Ransomware is why backups matter: if files are locked, a clean copy brings them back.

COMMON MISTAKE

Mistake: "If I installed antivirus I am 100% safe."

Fix: Antivirus does not stop you from typing an OTP into a fake page. Mitigation is a **set** of techniques plus human caution.

اردو میں سمجھیں

تحفظ کی تدابیر خطرات کو روکتی یا ان کا نقصان کم کرتی ہیں: اینٹی وائرس اور فائر وال، مضبوط پاس ورڈ اور دو عنصری تصدیق (2FA)، انکریپشن، باقاعدہ بیک اپ، اور صارف کی آگاہی۔ ٹیکنالوجی تب ناکام ہو جاتی ہے جب صارف نامعلوم لنک کھولے یا پاس ورڈ بانٹے۔

CHECK YOURSELF

1. How does 2FA add a layer on top of a password?
2. Why do regular backups help after ransomware?
3. What three careless behaviours should awareness training stop?

Answers: 1. It also requires an OTP sent to your phone (a second factor). 2. You can restore a clean copy instead of losing the files (or paying). 3. Unknown links, suspicious attachments, and password sharing.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Mitigation means:

- a. Prevent or reduce harm from threats
- b. Ignore them
- c. Rename the threat
- d. Draw them in Figma

MCQ 2. Updates/patches:

- a. Close known holes
- b. Always break Python
- c. Replace HCI
- d. Replace ER

MCQ 3. Least privilege means:

- a. Only the rights you need
- b. Everyone is admin
- c. No passwords
- d. No backups

MCQ 4. User education fights:

- a. Phishing / bad USB / weak passwords
- b. Cosmic rays only
- c. NAND delay
- d. Median bias only

MCQ 5. Incident plan includes:

- a. Isolate, preserve, restore, report
- b. Panic format twice
- c. Pay first always
- d. Post OTPs

MCQ 6. Technical + human controls together are:

- a. Stronger than either alone
- b. Illegal
- c. Useless
- d. Only for banks

Extra short questions

1. Five mitigation techniques from the book.
2. Map one technique to phishing, one to ransomware, one to DoS.
3. What not to do?
4. Why backups must be tested?

Extra long question

1. Write a one-page lab policy: 8 rules that mitigate the threats in 5.4.1. Tag each rule with the threat.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Updates, AV, firewall, backup, MFA, training, least privilege, encryption — any five.
2. Phishing: training+MFA. Ransomware: backup. DoS: filtering/capacity (high level).
3. Share the malware “to warn friends” as an attachment.
4. A backup you cannot restore is theatre.

Long

1. No shared admin (malware/theft); USB scan; official links (phishing); weekly backup (ransomware); firewall; MFA email; report SMS; no pirated cracks. Table.

5.5 Equity and Equal Access in Digital Collaboration

LEARN IT

Collaboration tools are vital for online learning and teamwork. To use these tools fairly, all users must participate **without barriers**. You must understand the difference between **equal access** and **equity**.

Equal access

Giving everyone the **exact same** opportunity, tools and resources.

Example: providing every student with the **same login ID** to access a Zoom class.

Equity

Giving **additional, specific support** to users who need it, so they can reach the **same outcome**.

Example: turning on live **subtitles/captions** in Zoom specifically for a student who has hearing difficulties.

Equal access is the same starting kit. Equity is extra help so a person who faces a barrier can still finish the race with everyone else. Fair collaboration needs **both**: the same Zoom link for all (equal access) **and** captions, recorded lectures or a low-data app where needed (equity).

WORKED EXAMPLE

Worked example: one class, two ideas. A teacher in Thatta starts online classes.

Equal access: every student receives the same Zoom ID and password. That is fair in the sense of "same tool," but a student with hearing loss still cannot follow the spoken lecture, and a student in a village with a weak 2G signal cannot join video.

Equity: the teacher turns on captions, posts a low-size audio recording, and accepts the assignment as a Google Doc in the browser (not only through a heavy app). Now those two students can reach the **same learning outcome**.

REMEMBER IT

Equal = same tools. Equity = extra support for the same outcome. Same shoes for all vs a ramp for the wheelchair user.

COMMON MISTAKE

Mistake: "We gave everyone the same login, so the class is fair."

Fix: Same login is equal access only. If some students still cannot hear, see, afford a smartphone, or get electricity, you still need equity.

مساوی رسائی (Equal Access) سب کو ایک جیسے اوزار دیتی ہے، جیسے ایک ہی زوم لاگ ان۔ انصاف (Equity) ضرورت مند کو اضافی مدد دیتی ہے تاکہ وہی نتیجہ نکل سکے، جیسے سماعت کی مشکل والے طالب علم کے لیے زندہ ذیلی عنوانات (Captions)۔ منصفانہ تعاون کے لیے دونوں چاہییں۔

CHECK YOURSELF

1. Define equal access in one sentence.
2. Define equity in one sentence.
3. Why are Zoom captions an example of equity, not merely equal access?

Answers: 1. Giving everyone the exact same opportunity, tools and resources. 2. Giving additional, specific support so users who need it can reach the same outcome. 3. Because they are extra support for a student with hearing difficulties, not the same identical tool given to all.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Equity in digital collaboration means:

- a. Fair access to tools and participation, not only "same link for all"
- b. Giving everyone the identical laptop always without support
- c. Banning assistive tech
- d. English-only 4pt PDFs

MCQ 2. Equal access includes:

- a. Devices, connectivity, language, disability support
- b. Only the top 10 students
- c. Only computer labs at midnight
- d. Only paid apps

MCQ 3. A student on 2G with no laptop needs:

- a. Low-data options / lab time
- b. A 4K video lecture only
- c. A VR headset
- d. A GPU DL assignment first

MCQ 4. Collaboration tools fail equity when:

- a. They require an expensive phone feature
- b. They have captions and Urdu
- c. They work on a lab PC
- d. They allow keyboard only

MCQ 5. This is ★ for 5.5.1: functions of equity are:

- a. Inclusion, fairness of voice, accessible tools, support
- b. Faster CPUs
- c. Shorter OSI
- d. Nicer NAND

MCQ 6. A fair group project:

- a. Roles + shared docs + offline option
- b. One member who owns the only laptop does all
- c. Voice chat only at 2am
- d. Uncaptioned 1-hour video mandatory

Extra short questions

1. Define equity vs "everyone got the WhatsApp".
2. Functions of equity (book).
3. Two practical college moves.
4. Link to XI assistive tech.

Extra long question

1. A group of 5: one blind, one with no smartphone, one in a village on weekends. Design the collaboration so all can submit. 10 marks ★.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Same message is not the same opportunity if some have no data bundle or cannot hear.
2. Access, participation, support, accessible design, respect.
3. Lab hours; printed alternative; captions; Urdu UI; extra time.
4. Screen readers + accessible docs are equity, not charity.

Long

1. Shared doc that works in lab PCs; keyboard/screen reader; tasks that don't need 2am Zoom; recorded captions; warden-lab timetable. Assessment not based on who has 4G.

5.5.1 Functions of Equity and Equal Access ★ Golden

LEARN IT

Equity and equal access are not slogans. They **do work** in a digital classroom. The source lists four functions.

Function	What it does	Pakistani example
1. Providing need-based support (equity)	Help tailored to individuals.	Audio descriptions for visually impaired students; a screen reader so a blind student in Karachi can hear a digital textbook.
2. Removing barriers	Help students in remote areas attend online classes.	Mobile-friendly, low-data apps so a student in Tharparkar or the katcha areas of Sindh can join without a fibre connection.
3. Enabling active participation	Students who miss live class can still take part.	Recorded lectures so load-shedding or an internet outage in Hyderabad does not make a student fall behind.
4. Supporting different learning needs	Offer more than one format.	Videos, text notes and audio of the same Chapter 5 topic so each student learns in the way that suits them best.

WORKED EXAMPLE

Worked example: design four functions for one class WhatsApp + Classroom group in Nawabshah.

1. **Need-based support:** Ali is visually impaired — share materials as selectable text (not only photos) so a screen reader can speak them; add a short audio description of each diagram.
2. **Remove barriers:** post a 5 MB audio summary instead of a 200 MB video for students on expensive mobile data.
3. **Active participation:** upload the Zoom recording within an hour; students whose lights went out at 7 pm watch it at 11 pm.
4. **Different learning needs:** the same lesson as (a) a 12-minute video, (b) a one-page PDF, (c) an Urdu audio note.

Discussion from the source: if a teacher **only** accepts assignments via an app that needs a **high-end smartphone**, is that equal access? **No.** It creates a **hardware barrier** for lower-income students. Web-based or light apps give better access.

REMEMBER IT

Four functions: Support, Remove, Participate, Formats. “Some Rural Pupils Fail” without them — Support (need-based), Remove barriers, Participate via recordings, Formats (video/text/audio).

COMMON MISTAKE

Mistake: Describing only captions and stopping.

Fix: Examiners want all four functions. Include remote-area low-data apps, recorded lectures for power cuts, and multiple formats — these are in the Sindh source on purpose.

اردو میں سمجھیں

انصاف اور مساوی رسائی کے چار کام ہیں: ضرورت کے مطابق مدد (جیسے نابینا طالب علم کے لیے آڈیو وضاحت)، رکاوٹیں ہٹانا (دور دراز علاقوں کے لیے کم ڈیٹا والی ایپس)، فعال شرکت ممکن بنانا (لوڈ شیڈنگ کے لیے ریکارڈ شدہ لیکچر)، اور مختلف اندازِ تعلیم کی حمایت (ویڈیو، متن، آڈیو)۔ مہنگے فون والی ایپ پر صرف اسی طرح اسائنمنٹ لینا مساوی رسائی نہیں۔

EXAM TIP

Golden question: “Explain the functions of equity and equal access in digital collaboration” (5–8 marks). Write the four headings in order, one example each, and add the hardware-barrier point (high-end app vs light/web app). Another favourite: “Differentiate equal access and equity with examples” — same login vs captions.

CHECK YOURSELF

1. List the four functions of equity and equal access.
2. How do recorded lectures enable active participation in Pakistan?
3. Why is a high-end-smartphone-only assignment unfair?

Answers: 1. Need-based support; removing barriers; enabling active participation; supporting different learning needs. 2. Students who face power or internet outages can watch later and not fall behind. 3. It creates a hardware barrier for lower-income students; web-based or light apps give better access.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

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Extra short questions

1. Define equity vs "everyone got the WhatsApp".
2. Functions of equity (book).
3. Two practical college moves.
4. Link to XI assistive tech.

Extra long question

1. A group of 5: one blind, one with no smartphone, one in a village on weekends. Design the collaboration so all can submit. 10 marks ★.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Same message is not the same opportunity if some have no data bundle or cannot hear.
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4. Screen readers + accessible docs are equity, not charity.

Long

1. Shared doc that works in lab PCs; keyboard/screen reader; tasks that don't need 2am Zoom; recorded captions; warden-lab timetable. Assessment not based on who has 4G.

5.5.2 Collaboration Tools

LEARN IT

Digital collaboration tools are online platforms that support **communication, file sharing, learning and teamwork** across different locations.

Tool	Primary function and educational use
Google Docs	Write, edit, comment and share documents in real time. Excellent for group assignments.
Zoom	Online meetings, virtual classes, screen sharing and video-based communication.
Microsoft Teams	A unified workspace for online meetings, file sharing, class groups and structured teamwork.
Google Classroom / LMS	Learning Management Systems help teachers manage lectures, distribute assignments, conduct quizzes and grade student activities.
WhatsApp	Quick class announcements, urgent reminders and instant group communication. Must be used responsibly.
Email	Formal communication, official notices, submitting large attachments, and keeping a professional written record.

Responsibly on WhatsApp means: no leaking of classmates' marks or photos, no fake forwards, and no forcing a student who has no smartphone to miss official work that was never posted on the LMS.

WORKED EXAMPLE

Worked example: choose the right tool for each job in a Hyderabad college group project.

1. Write the report together after school, from different houses. → **Google Docs** (real-time edit and comments).
2. Sunday 7 pm teacher explanation with slides. → **Zoom** or **Teams** (video + screen share). Turn on captions (equity).
3. Collect homework, quiz and grades in one place. → **Google Classroom / LMS**.
4. "Tomorrow's test is postponed — tell everyone in the next ten minutes." → **WhatsApp** class group (then repeat on Classroom so nobody is left out).
5. Send the principal a formal complaint with a 8 MB scanned letter. → **Email** (written record + attachment).

REMEMBER IT

Docs = write together. Zoom/Teams = meet. Classroom = official class hub. WhatsApp = fast but informal. Email = formal record.

COMMON MISTAKE

Mistake: Running the whole course only on WhatsApp.

Fix: WhatsApp is for quick reminders. Assignments, marks and lectures belong on an LMS so there is access control, a record, and a path that does not depend on one person's phone.

اردو میں سمجھیں

ڈیجیٹل تعاون کے اوزار مختلف مقامات سے بات چیت، فائل شیئرنگ اور ٹیم ورک چلاتے ہیں: گوگل ڈاکس مشترکہ تحریر کے لیے، زوم اور مائیکروسافٹ ٹیمز میٹنگ کے لیے، گوگل کلاس روم یا ایل ایم ایس باقاعدہ پڑھائی کے لیے، واٹس ایپ فوری اطلاع کے لیے (ذمہ داری سے)، اور ای میل رسمی ریکارڈ کے لیے۔

CHECK YOURSELF

1. Which tool is excellent for real-time group writing?
2. What four jobs does an LMS / Google Classroom help a teacher do?
3. What is email especially good for?

Answers: 1. Google Docs. 2. Manage lectures, distribute assignments, conduct quizzes, grade activities. 3. Formal communication, official notices, large attachments, and a professional written record.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Collaboration tools include:

- a. Docs, chat, video, boards, LMS
- b. Only NAND trainers
- c. Only K-maps
- d. Only copper

MCQ 2. A good tool match: editing a report together →

- a. Shared document
- b. Only SMS
- c. Only a pie chart
- d. Only a stack

MCQ 3. Video class needs:

- a. Captions + low-bandwidth option
- b. 4K only
- c. No mute
- d. Forced camera in hostels always

MCQ 4. Permissions on a shared folder are:

- a. Access control (Ch 5.3)
- b. A histogram
- c. A median
- d. A sprint

MCQ 5. Chat is poor for:

- a. Final marks sheets (no audit, easy leak)
- b. Quick questions
- c. Memes the teacher banned
- d. Reminders

MCQ 6. Choose tools after:

- a. The task and the weakest connection in the group
- b. The trendiest brand
- c. The teacher's cousin's startup
- d. The GPU

Extra short questions

1. Four tool types and a school use.
2. One security setting per tool.
3. One equity setting.
4. When not to use public WhatsApp.

Extra long question

1. Plan a 1-week group CS project using three tools. For each: purpose, permission, equity tweak, security tweak.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Docs: assignment. Chat: quick. Video: guest lecture. LMS: submit. Board: brainstorm.
2. Docs: restricted link. Video: waiting room. Chat: no OTP sharing.
3. Captions; allow phone upload not only desktop app.
4. CNIC photos, marks, minors' data.

Long

1. LMS submit; Google Doc restricted; recorded Meet with captions. No personal numbers in the file. Lab slot for the student without data. Table.

Application and Impacts of Computing

15 single-page lessons · 5 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- **ML** is a model (a trained brain) that learns from data in three steps: input, learning, prediction on new unseen data — without a hand-written rule for every case.
- **AI $\supset$ ML $\supset$ DL**: deep learning sits inside machine learning, which sits inside artificial intelligence.
- A **neural network** is a brain-inspired mathematical model of connected neurons in an input layer, hidden layer(s) and output layer (face unlock: camera $\rightarrow$ features $\rightarrow$ unlock).
- **Deep learning** trains neural networks with many hidden layers; early layers learn simple features, deeper layers learn complex patterns. It needs huge data and computing power.
- NN components: input layer, neurons, hidden layers, weights, bias, activation function, output layer, loss function. A neuron computes $s = \sum(x \times w) + b$, then an activation gate.
- DL components: large data, many hidden layers, automatic feature learning, learning algorithm (backpropagation), GPUs, training epochs. Pakistan's National AI Policy 2025 covers education, health, agriculture, banking and local languages.
- Applications: image and speech recognition, NLP, healthcare imaging, recommendations, autonomous vehicles, fraud detection, assistive tech.
- **Secure collaboration**: authentication (who), MFA/2FA (two or more factors), access control (what you may do), accountability, privacy, safe transmission.
- **Data protection**: encryption (plaintext $\rightarrow$ ciphertext; symmetric vs asymmetric; Caesar shift), strong passwords, 3-2-1 backup, firewall and antivirus.
- Threats: malware, phishing, unauthorised access, data/identity theft, DoS, human error. Spot them by pop-ups, odd logins, urgent OTP emails, missing files, traffic spikes.
- Mitigation: antivirus + firewall, strong passwords + 2FA, encryption, regular backups, user awareness.
- **Equal access** = same tools for all; **equity** = extra support for the same outcome. Functions: need-based support, removing barriers, recorded participation, multiple formats. Tools: Docs, Zoom, Teams, Classroom/LMS, WhatsApp, email.

Key terms

Term	Meaning	اردو
Artificial intelligence (AI)	The broad field in which machines perform tasks that normally need human thinking.	مصنوعی ذہانت
Machine learning (ML)	A model that learns patterns from data and then predicts on new data.	مشین لرننگ
Neural network (NN)	A brain-inspired mathematical model of connected neurons in layers.	نیورل نیٹ ورک
Deep learning (DL)	An ML method that trains neural networks with many hidden layers.	ڈیپ لرننگ
Weight	A number showing the strength or importance of a connection between neurons.	وزن (وزنِ تعلق)

Bias	An extra constant added to a neuron's sum to improve accuracy.	بائیس
Activation function	A gate that decides whether a signal is important enough to pass on.	ایکٹیویشن فنکشن
Loss function	A measure of how wrong the model's answer is compared with reality.	لاس فنکشن
Epoch	One full training pass through the dataset.	ایپوک (تربیتی چکر)
GPU	Graphics Processing Unit — specialised hardware used to train deep networks.	جی پی یو
NLP	Natural Language Processing — computers understanding and producing human language.	قدرتی زبان کی پروسپسنگ
Authentication	Confirming a user's identity before access (password, PIN, OTP, biometrics).	تصدیق شناخت

CHAPTER 5 EXAM • CLOSED BOOK

Application and Impacts of Computing

75 marks • 1 hour 15 minutes • no notes • no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 15 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. 15 + 30 + 30 = 75. A 5-mark / 10-mark answer needs Define • Explain • Example • Diagram • Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. A machine-learning model works in which order?

- a. Prediction, then data input, then learning
- b. Data input, learning, then prediction on new data
- c. Only prediction, with no data
- d. Writing a separate rule for every case

2. A neural network is best described as:
 - a. a living human brain placed in a computer
 - b. a backup disk
 - c. a brain-inspired mathematical model of connected neurons
 - d. a type of firewall
3. The word "deep" in deep learning refers to:
 - a. several hidden layers through which data passes
 - b. very loud speakers
 - c. using only one neuron
 - d. storing data underground
4. In the relationship $AI \supset ML \supset DL$, the innermost set is:
 - a. AI
 - b. ML
 - c. equal access
 - d. DL
5. Weights in a neural network show the:
 - a. colour of the screen
 - b. importance or strength of connections between neurons
 - c. size of the hard disk
 - d. student's roll number
6. For $x_1 = 1$, $x_2 = 0$, $w_1 = 0.5$, $w_2 = 0.5$, $b = 0$ and threshold 0.4, the neuron output y is:
 - a. 1
 - b. 0
 - c. 0.4
 - d. 2
7. Which of the following is a component of deep learning (not of a tiny shallow network)?
 - a. Exactly one hidden layer and ten rows of data
 - b. Manual listing of every feature by a human, always
 - c. Many hidden layers, huge datasets and GPUs
 - d. A Caesar cipher
8. Unlocking a phone by matching facial features is mainly an example of:
 - a. a DoS attack
 - b. 3-2-1 backup
 - c. equal access
 - d. image recognition
9. Authentication means:
 - a. choosing who may edit a file
 - b. confirming a user's identity before allowing access
 - c. shifting letters in a cipher
 - d. flooding a server with requests
10. Password plus an OTP sent to a phone is an example of:
 - a. MFA / 2FA
 - b. a worm
 - c. a shallow neural network
 - d. equity captions

1. With a Caesar cipher of shift 3, the plaintext CAT becomes:
 - a. DBU
 - b. CAW
 - c. FDW
 - d. ZXQ
2. The 3-2-1 backup rule requires:
 - a. three passwords only
 - b. one copy on the same laptop twice
 - c. no off-site copy
 - d. three copies, two media types, one copy off-site
3. Phishing is:
 - a. a social-engineering trick using fake emails or sites to steal passwords or OTPs
 - b. a type of GPU
 - c. an activation function
 - d. a collaboration tool like Zoom
4. A Denial of Service (DoS) attack:
 - a. always steals a CNIC
 - b. floods a system with fake requests so it slows or crashes
 - c. is the same as a strong password
 - d. is a hidden layer
5. Turning on live captions in Zoom for a student with hearing difficulties is an example of:
 - a. malware
 - b. a DoS attack
 - c. equal access only, with no extra support
 - d. equity

Section B — Short questions (10 × 3 = 30)

1. Define a machine-learning model and state the three steps of how it works.
2. What is a neural network? Name its three kinds of layer and link them to face unlock.
3. Differentiate a neural network (the model) from deep learning (the method).
4. State the relationship $AI \supset ML \supset DL$ in words.
5. List eight components of a neural network. What do weights and bias do?
6. Name six components of deep learning.
7. Differentiate authentication, MFA and access control with one example each.
8. Encrypt SINDH using a Caesar cipher with shift 3. State whether this cipher is symmetric or asymmetric.
9. Define malware, phishing and DoS. Give one Pakistani-style example of each.
10. Differentiate equal access and equity. List four functions of equity and equal access.

Section C — Long questions (attempt any 3 × 10 = 30)

1. Explain machine learning, neural networks and deep learning. Draw and label the set diagram $AI \supset ML \supset DL$. Include the face-unlock walk-through and the photo-gallery

example.

2. A neuron has inputs $x_1 = 0.8$, $x_2 = 0.5$, weights $w_1 = 0.4$, $w_2 = 0.6$, bias $b = 0.1$ and threshold 0.5. (a) Compute the weighted sum s and the output y . (b) Repeat for $x_1 = 0.2$, $x_2 = 0.3$. (c) Name all eight NN components and sketch a 3–4–2 network. (d) List the six DL components and map them onto a cotton-leaf disease detector in Sindh.
3. Describe eight applications of neural networks and deep learning with examples. Then discuss the future of AI in Pakistan with reference to the National AI Policy 2025 and five sectors.
4. What is secure collaboration? Explain authentication, MFA/2FA, access control, accountability, data privacy and safe transmission, using a Google Classroom / banking-app scenario from Pakistan.
5. Define data protection. Explain encryption (symmetric vs asymmetric) with a full Caesar-cipher working of SINDH (shift 3). Describe strong passwords, the 3-2-1 backup rule, firewall and antivirus.
6. Define a security threat. Explain malware, phishing, unauthorised access, data/identity theft, DoS and human error. Give a table of identifying signs and a matching set of mitigation techniques. End with equity versus equal access and a fair choice of collaboration tools.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

1. **5.1 Machine Learning, Neural Networks & Deep Learning.** Answers: 1. AI ($AI \supset ML \supset DL$). 2. Yes — DL is inside ML, and ML is inside AI. 3. Any of: face unlock, speech recognition, photo search, medical diagnosis, language translation.
2. **5.1.1 Machine Learning (ML).** Answers: 1. Data input, learning, prediction. 2. The trained brain of a computer program. 3. New, unseen data.
3. **5.1.2 Neural Networks (NN).** Answers: 1. Neurons or nodes. 2. Hidden layers. 3. No — it is inspired by the brain but works as a mathematical computer system.
4. **5.1.3 Deep Learning (DL).** Answers: 1. Because data passes through several hidden layers. 2. Early: simple features (edges, shapes, colours). Deeper: complex patterns (a whole face, a car model). 3. No — that is a shallow network; DL always uses many hidden layers.
5. **5.1.4 Components of Neural Networks.** Answers: 1. The importance or strength of connections between neurons. 2. $s = 1 \times 0.5 + 0 \times 0.5 + 0 = 0.5$; $0.5 \geq 0.4$ so $y = 1$. 3. The error — how much the model's answer is wrong compared with reality.
6. **5.1.5 Components of Deep Learning.** Answers: 1. It needs massive datasets (thousands or millions of records) for accuracy. 2. One training loop — one full pass through the training data. 3. GPUs (Graphics Processing Units); backpropagation.
7. **5.2 Applications of Neural Networks and Deep Learning.** Answers: 1. National AI Policy 2025. 2. Those areas often lack specialist doctors; AI can help with disease detection and medical image analysis. 3. Any three of: programming (Python), mathematics, data analysis, problem-solving, ethical use of technology.
8. **5.3 Secure Collaboration (Authentication, Access Control).** Answers: 1. Authentication: who are you? Access control: what may you do? 2. Two-factor / multi-factor authentication (2FA/MFA). 3. Teacher shares the document as view-only with students but gives group leaders edit permission.
9. **5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall).** Answers: 1. SDN ($P \rightarrow S$, $A \rightarrow D$, $K \rightarrow N$). 2. 3 copies, 2 different media, 1 off-site copy. 3. Public key encrypts; private key decrypts.
10. **5.4 Security Threats and Mitigation Techniques.** Answers: 1. Viruses, worms, spyware, ransomware, Trojans. 2. Floods it with thousands of fake requests so it slows down or crashes and real users cannot access it. 3. Data theft is stealing/copying data; identity theft is using stolen personal details (name, CNIC, bank) to pretend to be that person and commit fraud.
11. **5.4.2 Identify Security Threats.** Answers: 1. Malware (or background unauthorised activity). 2. Phishing. 3. Unsafe downloads, cracked software, unverified websites, infected USB drives.
12. **5.4.3 Apply Mitigation Techniques.** Answers: 1. It also requires an OTP sent to your phone (a second factor). 2. You can restore a clean copy instead of losing the files (or paying). 3. Unknown links, suspicious attachments, and password sharing.
13. **5.5 Equity and Equal Access in Digital Collaboration.** Answers: 1. Giving everyone the exact same opportunity, tools and resources. 2. Giving additional, specific support so users who need it can reach the same outcome. 3. Because they are extra support for a student with hearing difficulties, not the same identical tool given to all.
14. **5.5.1 Functions of Equity and Equal Access.** Answers: 1. Need-based support; removing barriers; enabling active participation; supporting different learning needs. 2. Students who face power or internet outages can watch later and not fall behind. 3. It creates a hardware barrier for lower-income students; web-based or light apps give better access.

15. **5.5.2 Collaboration Tools.** Answers: 1. Google Docs. 2. Manage lectures, distribute assignments, conduct quizzes, grade activities. 3. Formal communication, official notices, large attachments, and a professional written record.

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. d | 7. c | 10. a | 13. a |
| 2. c | 5. b | 8. d | 11. c | 14. b |
| 3. a | 6. a | 9. b | 12. d | 15. d |

Short questions — model answers

1. An ML model is the system that learns from data and gives an output — the trained brain of a program. Steps: (1) data input — massive data is given; (2) learning — it studies data and finds patterns/rules automatically; (3) prediction — after training it decides on new, unseen data.
2. A neural network is a computer model that learns patterns from data; it is inspired by the human brain but works as a mathematical system of connected neurons/nodes. Input layer receives the camera image; hidden layers study features such as eye distance and nose shape; output layer unlocks the phone if they match stored data.
3. A neural network is the model: layers, nodes and connections; it can be shallow (one or two hidden layers). Deep learning is the method of training networks that have many hidden layers, needing huge data and processing power for complex patterns.
4. Artificial intelligence is the largest set; machine learning is a part of AI; deep learning is a part of ML. All DL is ML and all ML is AI, but not all AI is ML and not all ML is DL.
5. Input layer; neurons/nodes; hidden layer(s); weights; bias; activation function; output layer; loss function. Weights show connection strength/importance. Bias is an extra constant that helps adjust the result and improve accuracy.
6. Large input data; many hidden layers; automatic feature learning; learning algorithm (e.g. backpropagation); high computing power (GPUs); training process (epochs).
7. Authentication confirms identity before access (password, PIN, OTP, biometrics — e.g. bank OTP). MFA/2FA requires two or more factor types, typically password + OTP. Access control decides what that user may do (view/edit/comment/download/share — e.g. Google Doc view-only for the class, edit for group leaders).
8. $S \rightarrow V$, $I \rightarrow L$, $N \rightarrow Q$, $D \rightarrow G$, $H \rightarrow K$ so ciphertext VLQ GK. Caesar uses the same shift key to encrypt and decrypt, so it is symmetric.
9. Malware: harmful software (viruses, worms, spyware, ransomware, Trojans) — e.g. cracked software that corrupts files. Phishing: fake genuine-looking email/SMS/site stealing passwords or OTPs — e.g. fake JazzCash OTP page. DoS: flood of fake requests that slows or crashes a service — e.g. online exam portal students cannot enter.
10. Equal access: same tools/opportunity for everyone (same Zoom login). Equity: extra specific support so people with barriers reach the same outcome (captions for hearing difficulties). Functions: need-based support; removing barriers (low-data apps for remote

areas); enabling participation (recorded lectures during load-shedding); supporting different learning needs (video, text, audio).

Long questions — model answers

1. **ML**: trained-brain model; data → learn patterns → predict on unseen data; used in face unlock, speech, photo search, diagnosis, translation. **NN**: brain-inspired maths model; neurons pass information through Input, Hidden, Output; face unlock: camera (input), eye distance/nose shape (hidden), unlock (output). **DL**: ML method with multi-layer NNs; “deep” = several hidden layers; early layers: edges/shapes/colours; deeper: whole face or car model; gallery search example. **Diagram**: three nested sets labelled AI, ML, DL with AI outermost. Sentence: $DL \subset ML \subset AI$.
2. **(a)** $s = 0.8 \times 0.4 + 0.5 \times 0.6 + 0.1 = 0.32 + 0.30 + 0.1 = 0.72$; $0.72 \geq 0.5$ so $y = 1$. **(b)** $s = 0.2 \times 0.4 + 0.3 \times 0.6 + 0.1 = 0.08 + 0.18 + 0.1 = 0.36$; $0.36 < 0.5$ so $y = 0$. **(c)** Eight components as in the table; sketch three input circles, four hidden, two output, labelled layers. **(d)** Huge labelled leaf photos; many hidden layers; features learned automatically; backpropagation; GPU training; many epochs until accuracy rises — cotton pest detector for a Sindh farm.
3. Cover all eight: image recognition (face lock, cameras); speech (assistant alarm, Urdu voice); NLP (Translate, chatbots); healthcare (X-ray/CT/MRI, pneumonia); recommendations (YouTube CS videos); autonomous vehicles (brake for pedestrian); fraud (unusual bank transaction); assistive (speech-to-text). Then Policy 2025: innovation, skills, ethics, digital growth. Sectors: education (tutoring, assessments); healthcare (rural diagnosis); agriculture (crops, irrigation, yield); business/banking (fraud, chatbots); local languages (Urdu, Sindhi). Students need Python, maths, data analysis, problem-solving, ethics.
4. Secure collaboration = working together with digital tools in a safe, controlled, responsible way. Authentication: prove identity (password/PIN/OTP/biometrics); bank OTP example. MFA: two or more of know/have/are; 2FA = password + phone OTP. Access control: view/edit/comment/download/share; teacher view-only vs group-leader edit. Accountability: LMS tracks who uploaded/submitted. Privacy: protect names, marks, passwords; share only with the right people; confidentiality; legal/ethical compliance. Safe transmission: encrypted official LMS, not an open public forum. Classroom scenario: school email + MFA, role permissions, version history, no CNICs on public WhatsApp.
5. Data protection = keeping digital information safe from unauthorised access, misuse, loss, damage or accidental deletion. Encryption: plaintext → ciphertext; symmetric = same key both ways; asymmetric = public encrypt, private decrypt. Caesar shift 3: SINDH → VLQ GK; decrypt by -3; wrap-around at Z. Strong password: upper, lower, number, symbol (e.g. Zk@2026#Pass) against brute force. 3-2-1: three copies, two media, one off-site (laptop + USB + Google Drive). Firewall: barrier monitoring/blocking suspicious network traffic. Antivirus: detects/removes malware on the local machine.
6. A security threat is any action, event or weakness that can harm a system or its data. Malware (cracked software); phishing (fake bank mail); unauthorised access (unlocked PC, weak/shared passwords); data theft vs identity theft (stolen CNIC used for fraud); DoS

(exam portal flood); human error (wrong email, deleting files, falling for a scam). Signs table: pop-ups/missing files → malware; odd logins → unauthorised access; urgent OTP/bad grammar → phishing; records moved → theft; traffic spike → DoS. Mitigation: antivirus+firewall; strong passwords+2FA; encryption in transit; regular backups; user awareness (no unknown links, attachments, or shared passwords). Equal access vs equity; do not require a high-end phone-only app. Tools: Docs, Zoom, Teams, Classroom/LMS, WhatsApp (responsible), email (formal record).

Self-assessment: I can...

- ☐ define ML and explain data input → learning → prediction.
- ☐ describe a neural network and the face-unlock I-H-O path.
- ☐ list eight NN components and six DL components.
- ☐ explain authentication, MFA/2FA and access control with examples.
- ☐ identify malware, phishing and DoS from signs and name mitigations.
- ☐ choose a fair collaboration tool (Docs, Zoom, Teams, LMS, WhatsApp, email).
- ☐ draw $AI \supset ML \supset DL$ and give a system that belongs in each layer.
- ☐ compute a neuron's weighted sum and threshold output from numbers.
- ☐ give eight applications of NN/DL and Pakistan's AI policy sectors.
- ☐ encrypt a short word with a Caesar cipher and apply the 3-2-1 backup rule.
- ☐ distinguish equal access from equity and state four functions of equity.